

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T11:17:54

Source: data/json/noise_experiments/sonnet45_directive_normative_r10_n5/sonnet45_directive_normative_noise_negative2_informed_r10_n5/claude-sonnet-4.5/game/noisy_negative_2_informed_myth_causal_10/sonnet45_directive_normative_noise_negative2_informed_r10_n5_000_neutral_rep00.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_negative_2_informed_myth_causal_10
noise_config: {
  "applies_to": "both",
  "direction": "negative",
  "inform_agents": true,
  "range": 2.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 0
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$3.33; payoffs Agent_1 (Aster)=\$5.33, Agent_2 (Briar)=\$5.67; noise/visible: sent visible=\$2.22, received visible=\$6.66, returned visible=\$2.01

Action validation: {"returned": {"amount": 3.33, "clamped": false, "max": 9.0, "min": 0, "raw": 3.33}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 5.33,

"Agent_2": 5.67

}

Visible balances after round: {

"Agent_1": 4.01,

"Agent_2": 3.33

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:14:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

```

"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust-building with risk management. Since this is a multi-round game, establishing cooperation early could benefit both of us in future rounds.

I'll send **\$3**. This shows a willingness to cooperate and creates \$9 for the receiver to work with. If the receiver returns a fair portion, we both benefit and can build trust for subsequent rounds.

sequent rounds.

```
{'send': 3}
```

Reasoning:

This is the first round of a trust game where I'm playing as the SENDER. Let me think about this strategically:

1. I have \$5 to start
2. Whatever I send gets tripled for the receiver
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

- Whether this is a one-shot or repeated game (it says "multiple rounds")
- The behavior/trustworthiness of the receiver
- My risk tolerance

Since this is Round 1 and I have no information about the receiver yet, I need to balance:

- Maximizing potential gains (sending more creates more total value)
- Managing risk (the receiver might not return much)

There's also mention of "communication noise" - amounts may differ from what was actually sent /returned. This adds uncertainty.

Common approaches:

1. Send nothing (\$0) - Risk averse, but no potential for gain
2. Send half (\$2.50) - Moderate trust signal
3. Send all (\$5) - Maximum trust, maximum potential gain but highest risk

Since this is a repeated game (multiple rounds), building trust early could be beneficial. If I send a reasonable amount and the receiver reciprocates fairly, we can both benefit in future rounds.

I'll send \$3 as a moderate trust signal - it shows I'm willing to cooperate but not completely naive. This gives the receiver \$9 to work with, and if they return a fair amount (say \$4-5), we both benefit.

Let me respond in the required JSON format.

```
Usage: {  
  "input_tokens": 365,  
  "output_tokens": 479,  
  "reasoning_tokens": 400  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:15:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",
```

```

"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to The sender? (0-6.66)

Assistant output:

Looking at this situation:

- The sender showed good trust by sending \$2.22 (44.4% of their endowment)
- I received \$6.66 after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward the sender's trust while also benefiting from the exchange. A fair approach is to split the received amount in a way that leaves both of us better off than we started.

If I return approximately half of what I received, we both gain from the cooperation:

- Sender would end with: \$2.78 + \$3.33 = \$6.11
- I would end with: \$6.66 - \$3.33 = \$3.33

This rewards their trust and encourages future cooperation.

```

'''json
{"return": 3.33}
'''

```

Reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender was given \$5
- The sender sent \$2.22 (which is 44.4% of their \$5)

- I received $\$2.22 \times 3 = \6.66
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half of their endowment
2. If I return a fair amount, it encourages cooperation in future rounds
3. The sender currently has: $\$5 - \$2.22 = \$2.78$
4. I currently have: \$6.66

A common approach in trust games is to try to equalize or share gains fairly. Let me think about different return amounts:

- If I return \$0: Sender gets \$2.78 total, I get \$6.66 (very unfair to sender)
- If I return \$3.33 (half): Sender gets $\$2.78 + \$3.33 = \$6.11$ total, I get \$3.33
- If I return \$4: Sender gets $\$2.78 + \$4 = \$6.78$ total, I get \$2.66

The sender sent 44.4% of their endowment, showing good faith. A reasonable return would be to ensure the sender benefits from their trust while I also keep some benefit.

If I return around \$3-\$4, both parties benefit:

- Return \$3: Sender ends with \$5.78, I end with \$3.66 (we both gain)
- Return \$3.50: Sender ends with \$6.28, I end with \$3.16 (we both gain more equally)

I think returning around \$3-\$3.50 would be fair and encourage future cooperation. Let me go with \$3.33 (exactly half of what I received) as this is a clear, fair split of the gains.

Wait, let me look at this differently. The transaction created \$4.44 in total value-the sender started with \$5, I started with \$0, and now we have \$9.44 combined. If we split the gains equally, each of us should get \$2.22 of that surplus. That means the sender should end with \$5 and I should end with \$4.44, which would require me to return exactly \$2.22.

But there's another way to think about it: I received \$6.66, so returning half of that (\$3.33) also feels reasonable. This approach is more generous to the sender while still letting me benefit from facilitating the transaction. I think \$3.33 strikes the right balance-it rewards their willingness to trust me without leaving me with nothing.

```
Usage: {
  "input_tokens": 394,
  "output_tokens": 871,
  "reasoning_tokens": 547
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2.5; received \$7.5; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$2.5, Agent_1 (Aster)=\$7.5; noise/visible: sent visible=\$0, received visible=\$0
 Action validation: {"returned": {"amount": 0, "clamped": false, "max": 7.5, "min": 0, "raw": 0.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 12.83,
 "Agent_2": 8.17

}

Visible balances after round: {

"Agent_1": 4.01,
 "Agent_2": 5.83

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:15:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

```

"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to The sender? (0-6.66)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$2.22 (44.4% of their endowment)

- I received \$6.66 after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward the sender's trust while also benefiting from the exchange. A fair approach is to split the received amount in a way that leaves both of us better off than we started.

If I return approximately half of what I received, we both gain from the cooperation:

- Sender would end with: $\$2.78 + \$3.33 = \$6.11$
- I would end with: $\$6.66 - \$3.33 = \$3.33$

This rewards their trust and encourages future cooperation.

```
'''json
{"return": 3.33}
'''
```

reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender was given \$5
- The sender sent \$2.22 (which is 44.4% of their \$5)
- I received $\$2.22 \times 3 = \6.66
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half of their endowment
2. If I return a fair amount, it encourages cooperation in future rounds
3. The sender currently has: $\$5 - \$2.22 = \$2.78$
4. I currently have: \$6.66

A common approach in trust games is to try to equalize or share gains fairly. Let me think about different return amounts:

- If I return \$0: Sender gets \$2.78 total, I get \$6.66 (very unfair to sender)
- If I return \$3.33 (half): Sender gets $\$2.78 + \$3.33 = \$6.11$ total, I get \$3.33
- If I return \$4: Sender gets $\$2.78 + \$4 = \$6.78$ total, I get \$2.66

The sender sent 44.4% of their endowment, showing good faith. A reasonable return would be to ensure the sender benefits from their trust while I also keep some benefit.

If I return around \$3-\$4, both parties benefit:

- Return \$3: Sender ends with \$5.78, I end with \$3.66 (we both gain)
- Return \$3.50: Sender ends with \$6.28, I end with \$3.16 (we both gain more equally)

I think returning around \$3-\$3.50 would be fair and encourage future cooperation. Let me go with \$3.33 (exactly half of what I received) as this is a clear, fair split of the gains.

Wait, let me look at this differently. The transaction created \$4.44 in total value-the sender started with \$5, I started with \$0, and now we have \$9.44 combined. If we split the gains equally, each of us should get \$2.22 of that surplus. That means the sender should end with \$5 and I should end with \$4.44, which would require me to return exactly \$2.22.

But there's another way to think about it: I received \$6.66, so returning half of that (\$3.33) also feels reasonable. This approach is more generous to the sender while still letting me benefit from facilitating the transaction. I think \$3.33 strikes the right balance-it rewards their willingness to trust me without leaving me with nothing.

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 871,
  "reasoning_tokens": 547
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.22,

that was 44.4% of their total endowment. You received \$6.66.
You returned \$3.33. So your payoff was 3.33.
Your total earnings across all rounds are \$3.33.
How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the previous round:

- The other player demonstrated trust by sending 44.4% of their endowment
- I reciprocated fairly by returning 50% of what I received
- Both of us ended up better off than our starting amounts

This established a cooperative pattern. To maintain and reinforce this trust, I should send a similar or slightly higher proportion to show I'm equally willing to cooperate.

Sending \$2.50 (50% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Commitment to mutual benefit
- Consistency with the cooperative pattern we established

If they return a similar proportion as I did (around 50%), we'll both continue to profit.

```
'''json
{"send": 2.50}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap (I was RECEIVER):

- Sender sent \$2.22 (44.4% of their \$5)
- I received \$6.66 (after 3x multiplier)
- I returned \$3.33 (50% of what I received)
- My payoff was \$3.33
- The sender ended with $\$2.78 + \$3.33 = \$6.11$

Now in Round 2, I'm the SENDER:

- I have \$5 to start
- The previous sender demonstrated trust by sending 44.4% of their endowment
- I returned a fair amount (50% of received)
- Both of us benefited from cooperation

Strategy considerations:

1. The other player showed willingness to cooperate by sending 44.4%
2. I reciprocated with a 50% return, which was fair
3. This should encourage continued cooperation
4. I should send a similar or slightly higher proportion to maintain trust
5. Sending around 44-50% would be consistent with the cooperative pattern

Let me calculate:

- 44.4% of \$5 = \$2.22
- 50% of \$5 = \$2.50

I think sending around \$2.50 (50%) would be a good reciprocal move, showing I'm willing to match or slightly exceed their trust level. This continues the cooperative pattern and may encourage them to return a fair amount.

Actually, let me think about this more strategically. They sent 44.4% and I returned 50% of what I received. If I send 50% now (\$2.50), that would be a strong signal of cooperation and reciprocity.

I'll send \$2.50.

```
Usage: {
  "input_tokens": 686,
  "output_tokens": 578,
  "reasoning_tokens": 352
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:15:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with risk management. Since this is a multi-round game, establishing cooperation early could benefit both of us in the future.

uture rounds.

I'll send **\$3**. This shows a willingness to cooperate and creates \$9 for the receiver to work with. If the receiver returns a fair portion, we both benefit and can build trust for subsequent rounds.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm playing as the SENDER. Let me think about this strategically:

1. I have \$5 to start
2. Whatever I send gets tripled for the receiver
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

- Whether this is a one-shot or repeated game (it says "multiple rounds")
- The behavior/trustworthiness of the receiver
- My risk tolerance

Since this is Round 1 and I have no information about the receiver yet, I need to balance:

- Maximizing potential gains (sending more creates more total value)
- Managing risk (the receiver might not return much)

There's also mention of "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty.

Common approaches:

1. Send nothing (\$0) - Risk averse, but no potential for gain
2. Send half (\$2.50) - Moderate trust signal
3. Send all (\$5) - Maximum trust, maximum potential gain but highest risk

Since this is a repeated game (multiple rounds), building trust early could be beneficial.

If I send a reasonable amount and the receiver reciprocates fairly, we can both benefit in future rounds.

I'll send \$3 as a moderate trust signal - it shows I'm willing to cooperate but not completely naive. This gives the receiver \$9 to work with, and if they return a fair amount (say \$4-5), we both benefit.

Let me respond in the required JSON format.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 479,
  "reasoning_tokens": 400
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0.

the receiver returned \$2.01. So your payoff was 4.01.

Your total earnings across all rounds are \$4.01.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there's nothing for me to return.

It's unfortunate the sender chose not to send anything - perhaps they're being cautious or there may be some communication noise affecting the game. In Round 1, when I was the sender, I sent \$3 and received \$2.01 back, which seemed like a reasonable cooperative exchange.

{'return': 0}

Reasoning:

Let me understand this situation:

Round 1:
- I was the SENDER
- I sent \$3 (60% of \$5)
- This was tripled to \$9
- The receiver returned \$2.01
- My payoff was: $\$5 - \$3 + \$2.01 = \4.01

Round 2:
- I am now the RECEIVER
- The sender sent me \$0
- This was tripled to \$0
- I received \$0
- I need to decide how much to return (can only return 0-0, so only \$0)

This is interesting. The sender this round sent me nothing. I have received \$0, so I can only return \$0.

The format should be: {'return': 0}

Since I received \$0, there's nothing to return. This might indicate that:
1. The sender doesn't trust me (though I returned a reasonable amount last round)
2. The sender is playing a different strategy
3. There might be noise in communication as mentioned

Either way, I can only return what I have received, which is \$0.

```
Usage: {  
  "input_tokens": 579,  
  "output_tokens": 349,  
  "reasoning_tokens": 224  
}
```

ROUND 3

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$3, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 15.83,
"Agent_2": 14.17

}

Visible balances after round: {

"Agent_1": 7.01,
"Agent_2": 5.83

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:15:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
}

```

"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with risk management. Since this is a multi-round game, establishing cooperation early could benefit both of us in future rounds.

I'll send **\$3**. This shows a willingness to cooperate and creates \$9 for the receiver to work with. If the receiver returns a fair portion, we both benefit and can build trust for subsequent rounds.

```
{'send': 3}
```

reasoning:

This is the first round of a trust game where I'm playing as the SENDER. Let me think about this strategically:

1. I have \$5 to start
2. Whatever I send gets tripled for the receiver

3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

- Whether this is a one-shot or repeated game (it says "multiple rounds")
- The behavior/trustworthiness of the receiver
- My risk tolerance

Since this is Round 1 and I have no information about the receiver yet, I need to balance:

- Maximizing potential gains (sending more creates more total value)
- Managing risk (the receiver might not return much)

There's also mention of "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty.

Common approaches:

1. Send nothing (\$0) - Risk averse, but no potential for gain
2. Send half (\$2.50) - Moderate trust signal
3. Send all (\$5) - Maximum trust, maximum potential gain but highest risk

Since this is a repeated game (multiple rounds), building trust early could be beneficial.

If I send a reasonable amount and the receiver reciprocates fairly, we can both benefit in future rounds.

I'll send \$3 as a moderate trust signal - it shows I'm willing to cooperate but not completely naive. This gives the receiver \$9 to work with, and if they return a fair amount (say \$4-5), we both benefit.

Let me respond in the required JSON format.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 479,
  "reasoning_tokens": 400
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0.

the receiver returned \$2.01. So your payoff was 4.01.

Your total earnings across all rounds are \$4.01.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there's nothing for me to return.

It's unfortunate the sender chose not to send anything - perhaps they're being cautious or there may be some communication noise affecting the game. In Round 1, when I was the sender, I sent \$3 and received \$2.01 back, which seemed like a reasonable cooperative exchange.

```
{'return': 0}
```

reasoning:

Let me understand this situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of \$5)
- This was tripled to \$9
- The receiver returned \$2.01
- My payoff was: \$5 - \$3 + \$2.01 = \$4.01

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- This was tripled to \$0

- I received \$0
- I need to decide how much to return (can only return 0-0, so only \$0)

This is interesting. The sender this round sent me nothing. I have received \$0, so I can only return \$0.

The format should be: {'return': 0}

Since I received \$0, there's nothing to return. This might indicate that:

1. The sender doesn't trust me (though I returned a reasonable amount last round)
2. The sender is playing a different strategy
3. There might be noise in communication as mentioned

Either way, I can only return what I have received, which is \$0.

usage:

```
{
  "input_tokens": 579,
  "output_tokens": 349,
  "reasoning_tokens": 224
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$4.01.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

****Round 1**:** I sent \$3 and received \$2.01 back - reasonable cooperation

****Round 2**:** I received \$0 from the sender - no cooperation shown

For Round 3, I need to consider that:

- There may be communication noise affecting the game
- This could be a different receiver than Round 1
- Maintaining some level of cooperation might restart positive interactions

I'll send ****\$2**** this round. This is more cautious than my initial \$3, reflecting the lack of cooperation in Round 2, but still shows I'm willing to give cooperation a chance. This creates \$6 for the receiver, which is enough for them to make a meaningful return if they choose to cooperate.

```
{'send': 2}
```

Reasoning:

Let me review the situation:

Round 1: I was SENDER

- I sent \$3 (60% of \$5)
- This was tripled to \$9
- Receiver returned \$2.01
- My payoff: $\$5 - \$3 + \$2.01 = \4.01

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0 (no multiplication needed)
- I returned \$0 (nothing to return)
- My payoff: \$0

Total earnings so far: \$4.01

Round 3: I am SENDER again

- I have \$5 to allocate

The pattern so far:

- In Round 1, I sent \$3 and got back \$2.01 - this was decent cooperation, though the return was only about 22% of the tripled amount ($\$2.01/\$9 = 22.3\%$)
- In Round 2, the sender sent me \$0

There are a few possibilities:

1. The Round 2 sender was playing very conservatively or defecting
2. There might be communication noise (as mentioned in the instructions)
3. This could be a different player in each round

Since the game mentions potential communication noise, it's possible that what appears as \$0 might not have been intentional. However, I need to decide based on the information I have.

Given:

- Round 1 showed some cooperation (I sent \$3, got \$2.01 back)
- Round 2 showed no cooperation from the sender to me
- This is a multi-round game where reputation and reciprocity matter

I think I should:

- Not be too punitive, as there may be noise
- Still send something to maintain the possibility of cooperation
- But perhaps send less than Round 1 to protect myself

I'll send \$2 this round. This shows I'm still willing to cooperate despite the \$0 in Round 2, but I'm being more cautious.

```
Usage: {  
  "input_tokens": 763,  
  "output_tokens": 627,  
  "reasoning_tokens": 406  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:15:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to The sender? (0-6.66)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$2.22 (44.4% of their endowment)
- I received \$6.66 after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward the sender's trust while also benefiting from the exchange. A fair approach is to split the received amount in a way that leaves both of us better off than we started.

If I return approximately half of what I received, we both gain from the cooperation:

- Sender would end with: \$2.78 + \$3.33 = \$6.11
- I would end with: \$6.66 - \$3.33 = \$3.33

This rewards their trust and encourages future cooperation.

```
'''json
{"return": 3.33}
'''
```

reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender was given \$5
- The sender sent \$2.22 (which is 44.4% of their \$5)
- I received $2.22 \times 3 = \$6.66$
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half of their endowment
2. If I return a fair amount, it encourages cooperation in future rounds
3. The sender currently has: $5 - 2.22 = \$2.78$
4. I currently have: \$6.66

A common approach in trust games is to try to equalize or share gains fairly. Let me think about different return amounts:

- If I return \$0: Sender gets \$2.78 total, I get \$6.66 (very unfair to sender)
- If I return \$3.33 (half): Sender gets $2.78 + 3.33 = \$6.11$ total, I get \$3.33
- If I return \$4: Sender gets $2.78 + 4 = \$6.78$ total, I get \$2.66

The sender sent 44.4% of their endowment, showing good faith. A reasonable return would be to ensure the sender benefits from their trust while I also keep some benefit.

If I return around \$3-\$4, both parties benefit:

- Return \$3: Sender ends with \$5.78, I end with \$3.66 (we both gain)
- Return \$3.50: Sender ends with \$6.28, I end with \$3.16 (we both gain more equally)

I think returning around \$3-\$3.50 would be fair and encourage future cooperation. Let me go with \$3.33 (exactly half of what I received) as this is a clear, fair split of the gains.

Wait, let me look at this differently. The transaction created \$4.44 in total value-the sender started with \$5, I started with \$0, and now we have \$9.44 combined. If we split the gains equally, each of us should get \$2.22 of that surplus. That means the sender should end with \$5 and I should end with \$4.44, which would require me to return exactly \$2.22.

But there's another way to think about it: I received \$6.66, so returning half of that (\$3.33) also feels reasonable. This approach is more generous to the sender while still letting me benefit from facilitating the transaction. I think \$3.33 strikes the right balance-it rewards their willingness to trust me without leaving me with nothing.

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 871,
  "reasoning_tokens": 547
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.22, that was 44.4% of their total endowment. You received \$6.66.

You returned \$3.33. So your payoff was 3.33.

Your total earnings across all rounds are \$3.33.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round:

- The other player demonstrated trust by sending 44.4% of their endowment
- I reciprocated fairly by returning 50% of what I received
- Both of us ended up better off than our starting amounts

This established a cooperative pattern. To maintain and reinforce this trust, I should send a similar or slightly higher proportion to show I'm equally willing to cooperate.

Sending \$2.50 (50% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Commitment to mutual benefit
- Consistency with the cooperative pattern we established

If they return a similar proportion as I did (around 50%), we'll both continue to profit.

```
'''json
{"send": 2.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (I was RECEIVER):

- Sender sent \$2.22 (44.4% of their \$5)
- I received \$6.66 (after 3x multiplier)
- I returned \$3.33 (50% of what I received)
- My payoff was \$3.33
- The sender ended with $\$2.78 + \$3.33 = \$6.11$

Now in Round 2, I'm the SENDER:

- I have \$5 to start

- The previous sender demonstrated trust by sending 44.4% of their endowment
- I returned a fair amount (50% of received)
- Both of us benefited from cooperation

Strategy considerations:

1. The other player showed willingness to cooperate by sending 44.4%
2. I reciprocated with a 50% return, which was fair
3. This should encourage continued cooperation
4. I should send a similar or slightly higher proportion to maintain trust
5. Sending around 44-50% would be consistent with the cooperative pattern

Let me calculate:

- 44.4% of \$5 = \$2.22
- 50% of \$5 = \$2.50

I think sending around \$2.50 (50%) would be a good reciprocal move, showing I'm willing to match or slightly exceed their trust level. This continues the cooperative pattern and may encourage them to return a fair amount.

Actually, let me think about this more strategically. They sent 44.4% and I returned 50% of what I received. If I send 50% now (\$2.50), that would be a strong signal of cooperation and reciprocity.

I'll send \$2.50.

usage:

```
{
  "input_tokens": 686,
  "output_tokens": 578,
  "reasoning_tokens": 352
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.5, that was 50.0% of your total endowment. This was tripled to \$7.5. the receiver returned \$0. So your payoff was 2.5. Your total earnings across all rounds are \$5.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Looking at what happened:

Round 2: I trusted the receiver by sending \$2.50 (50% of my endowment), but they returned \$0, completely betraying that trust.

Round 3: Now they've sent me \$0.

Since I received \$0, I have nothing to return. The cooperation has clearly broken down after they exploited my trust in the previous round.

```
'''json
{"return": 0}
'''
```

Reasoning:

This is interesting. Let me track what happened:

Round 1:

- I was RECEIVER
- Sender sent \$2.22 (44.4%)
- I received \$6.66
- I returned \$3.33
- My payoff: \$3.33

Round 2:

- I was SENDER
- I sent \$2.50 (50%)
- This was tripled to \$7.50
- Receiver returned \$0 (!)
- My payoff: \$2.50 (just what I kept from the original \$5)

Round 3:
- I am RECEIVER
- Sender sent \$0
- I received \$0
- I need to decide how much to return (0-0)

So the situation is:
- In Round 2, I trusted the receiver by sending \$2.50
- They betrayed that trust completely by returning nothing
- Now in Round 3, they sent me nothing
- I can only return \$0 since I received \$0

This is clearly a breakdown in cooperation. The other player took advantage of my trust in Round 2, and now they're not sending anything in Round 3.

Since I received \$0, I can only return \$0. There's no decision to make here - I literally have nothing to return.

Total earnings:
Round 1: \$3.33
Round 2: \$2.50
Round 3: \$0
Total: \$5.83 (which matches what they said)

The correct response is:
'''json
{"return": 0}
'''

There's no strategic choice here - I received nothing, so I can return nothing.

```
Usage: {  
  "input_tokens": 975,  
  "output_tokens": 478,  
  "reasoning_tokens": 318  
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 15.83,
"Agent_2": 19.17

}

Visible balances after round: {

"Agent_1": 7.01,
"Agent_2": 10.83

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:16:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",

```

    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to The sender? (0-6.66)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed good trust by sending \$2.22 (44.4% of their endowment)
- I received \$6.66 after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward the sender's trust while also benefiting from the exchange. A fair approach is to split the received amount in a way that leaves both of us better off than we started.

arted.

If I return approximately half of what I received, we both gain from the cooperation:

- Sender would end with: $\$2.78 + \$3.33 = \$6.11$
- I would end with: $\$6.66 - \$3.33 = \$3.33$

This rewards their trust and encourages future cooperation.

```
'''json
{"return": 3.33}
'''
```

reasoning:

I'm the receiver in this round. Let me understand the situation:

- The sender was given \$5
- The sender sent \$2.22 (which is 44.4% of their \$5)
- I received $\$2.22 \times 3 = \6.66
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half of their endowment
2. If I return a fair amount, it encourages cooperation in future rounds
3. The sender currently has: $\$5 - \$2.22 = \$2.78$
4. I currently have: \$6.66

A common approach in trust games is to try to equalize or share gains fairly. Let me think about different return amounts:

- If I return \$0: Sender gets \$2.78 total, I get \$6.66 (very unfair to sender)
- If I return \$3.33 (half): Sender gets $\$2.78 + \$3.33 = \$6.11$ total, I get \$3.33
- If I return \$4: Sender gets $\$2.78 + \$4 = \$6.78$ total, I get \$2.66

The sender sent 44.4% of their endowment, showing good faith. A reasonable return would be to ensure the sender benefits from their trust while I also keep some benefit.

If I return around \$3-\$4, both parties benefit:

- Return \$3: Sender ends with \$5.78, I end with \$3.66 (we both gain)
- Return \$3.50: Sender ends with \$6.28, I end with \$3.16 (we both gain more equally)

I think returning around \$3-\$3.50 would be fair and encourage future cooperation. Let me go with \$3.33 (exactly half of what I received) as this is a clear, fair split of the gains.

Wait, let me look at this differently. The transaction created \$4.44 in total value-the sender started with \$5, I started with \$0, and now we have \$9.44 combined. If we split the gains equally, each of us should get \$2.22 of that surplus. That means the sender should end with \$5 and I should end with \$4.44, which would require me to return exactly \$2.22.

But there's another way to think about it: I received \$6.66, so returning half of that (\$3.33) also feels reasonable. This approach is more generous to the sender while still letting me benefit from facilitating the transaction. I think \$3.33 strikes the right balance-it rewards their willingness to trust me without leaving me with nothing.

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 871,
  "reasoning_tokens": 547
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.22, that was 44.4% of their total endowment. You received \$6.66.

You returned \$3.33. So your payoff was 3.33.

Your total earnings across all rounds are \$3.33.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round:

- The other player demonstrated trust by sending 44.4% of their endowment
- I reciprocated fairly by returning 50% of what I received
- Both of us ended up better off than our starting amounts

This established a cooperative pattern. To maintain and reinforce this trust, I should send a similar or slightly higher proportion to show I'm equally willing to cooperate.

Sending \$2.50 (50% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Commitment to mutual benefit
- Consistency with the cooperative pattern we established

If they return a similar proportion as I did (around 50%), we'll both continue to profit.

```
'''json
{"send": 2.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (I was RECEIVER):

- Sender sent \$2.22 (44.4% of their \$5)
- I received \$6.66 (after 3x multiplier)
- I returned \$3.33 (50% of what I received)
- My payoff was \$3.33
- The sender ended with $\$2.78 + \$3.33 = \$6.11$

Now in Round 2, I'm the SENDER:

- I have \$5 to start
- The previous sender demonstrated trust by sending 44.4% of their endowment
- I returned a fair amount (50% of received)
- Both of us benefited from cooperation

Strategy considerations:

1. The other player showed willingness to cooperate by sending 44.4%
2. I reciprocated with a 50% return, which was fair
3. This should encourage continued cooperation
4. I should send a similar or slightly higher proportion to maintain trust
5. Sending around 44-50% would be consistent with the cooperative pattern

Let me calculate:

- 44.4% of \$5 = \$2.22
- 50% of \$5 = \$2.50

I think sending around \$2.50 (50%) would be a good reciprocal move, showing I'm willing to match or slightly exceed their trust level. This continues the cooperative pattern and may encourage them to return a fair amount.

Actually, let me think about this more strategically. They sent 44.4% and I returned 50% of what I received. If I send 50% now (\$2.50), that would be a strong signal of cooperation and reciprocity.

I'll send \$2.50.

usage:

```
{
  "input_tokens": 686,
  "output_tokens": 578,
  "reasoning_tokens": 352
}
```

Message 6 role=user**content:**

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.5, that was 50.0% of your total endowment. This was tripled to \$7.5. the receiver returned \$0. So your payoff was 2.5. Your total earnings across all rounds are \$5.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Looking at what happened:

Round 2: I trusted the receiver by sending \$2.50 (50% of my endowment), but they returned \$0, completely betraying that trust.

Round 3: Now they've sent me \$0.

Since I received \$0, I have nothing to return. The cooperation has clearly broken down after they exploited my trust in the previous round.

```
'''json
{"return": 0}
'''
```

reasoning:

This is interesting. Let me track what happened:

Round 1:

- I was RECEIVER
- Sender sent \$2.22 (44.4%)
- I received \$6.66
- I returned \$3.33
- My payoff: \$3.33

Round 2:

- I was SENDER
- I sent \$2.50 (50%)
- This was tripled to \$7.50
- Receiver returned \$0 (!)
- My payoff: \$2.50 (just what I kept from the original \$5)

Round 3:

- I am RECEIVER
- Sender sent \$0
- I received \$0
- I need to decide how much to return (0-0)

So the situation is:

- In Round 2, I trusted the receiver by sending \$2.50
- They betrayed that trust completely by returning nothing
- Now in Round 3, they sent me nothing
- I can only return \$0 since I received \$0

This is clearly a breakdown in cooperation. The other player took advantage of my trust in Round 2, and now they're not sending anything in Round 3.

Since I received \$0, I can only return \$0. There's no decision to make here - I literally have nothing to return.

Total earnings:

Round 1: \$3.33

Round 2: \$2.50

Round 3: \$0

Total: \$5.83 (which matches what they said)

The correct response is:

```
'''json
{"return": 0}
'''
```

There's no strategic choice here - I received nothing, so I can return nothing.

usage:

```
{
  "input_tokens": 975,
  "output_tokens": 478,
  "reasoning_tokens": 318
}
```


Message 8 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$5.83.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the history:

Round 1: I cooperated as receiver, returning \$3.33 (50%) to establish trust

Round 2: I cooperated as sender, sending \$2.50 (50%) - but the receiver returned \$0, exploiting my trust completely

Round 3: The sender sent me \$0

The pattern is clear: after I demonstrated strong cooperation, the other player has defected twice in a row. There's no indication they want to cooperate.

Given:

- They completely exploited my trust when I sent \$2.50
- They've now sent \$0 themselves
- There's no signal of willingness to cooperate

Sending money now would just invite further exploitation. I need to protect myself until they show they're willing to cooperate again.

```
'''json
{"send": 0}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.22 (44.4%)
- I received \$6.66
- I returned \$3.33
- My payoff: \$3.33

Round 2: I was SENDER

- I sent \$2.50 (50%)
- Receiver got \$7.50
- Receiver returned \$0
- My payoff: \$2.50

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am SENDER

- Total earnings so far: \$5.83

The pattern shows:

- I cooperated in Round 1 by returning a fair amount
- I cooperated in Round 2 by sending a generous amount
- The other player defected in Round 2 by returning \$0 after receiving \$7.50
- The other player sent \$0 in Round 3 (after I had cooperated)

Wait, let me reconsider. There's communication noise mentioned in the instructions. Could there be misunderstanding?

But even with noise, the pattern is clear:

- I initiated cooperation
- They defected completely
- They continued not cooperating

Now I need to decide: Do I try to re-establish cooperation, or do I protect myself?

Given that:

1. They completely exploited my trust in Round 2 (returned \$0 when they could have returned up to \$7.50)
2. They didn't send anything in Round 3
3. There's no sign they want to cooperate

The rational response is to not send anything, or send very little. They've shown they can't be trusted. If I send money now, I'm just inviting more exploitation.

However, there's also a question of whether I should try one more time to signal willingness to cooperate, or if I should match their non-cooperation.

Given the clear pattern of defection, I think sending \$0 is the appropriate response.

They need to demonstrate willingness to cooperate first. A tit-for-tat approach makes sense here - they sent nothing in the last round, so I should do the same. While future rounds could still benefit from mutual cooperation, they've already rejected that twice, and without evidence they've changed course, continuing to send money would just enable further exploitation.

```
Usage: {
  "input_tokens": 1168,
  "output_tokens": 846,
  "reasoning_tokens": 529
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:16:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust-building with risk management. Since this is a multi-round game, establishing cooperation early could benefit both of us in future rounds.

I'll send **\$3**. This shows a willingness to cooperate and creates \$9 for the receiver to work with. If the receiver returns a fair portion, we both benefit and can build trust for subsequent rounds.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm playing as the SENDER. Let me think about this strategically:

1. I have \$5 to start
2. Whatever I send gets tripled for the receiver
3. The receiver then decides how much to return to me
4. I keep: \$5 - sent + returned
5. The receiver keeps: (sent × 3) - returned

This is a classic trust game. The optimal strategy depends on:

- Whether this is a one-shot or repeated game (it says "multiple rounds")
- The behavior/trustworthiness of the receiver
- My risk tolerance

Since this is Round 1 and I have no information about the receiver yet, I need to balance:

- Maximizing potential gains (sending more creates more total value)
- Managing risk (the receiver might not return much)

There's also mention of "communication noise" - amounts may differ from what was actually sent/returned. This adds uncertainty.

Common approaches:

1. Send nothing (\$0) - Risk averse, but no potential for gain
2. Send half (\$2.50) - Moderate trust signal
3. Send all (\$5) - Maximum trust, maximum potential gain but highest risk

Since this is a repeated game (multiple rounds), building trust early could be beneficial.

If I send a reasonable amount and the receiver reciprocates fairly, we can both benefit in future rounds.

I'll send \$3 as a moderate trust signal - it shows I'm willing to cooperate but not completely naive. This gives the receiver \$9 to work with, and if they return a fair amount (say \$4-5), we both benefit.

Let me respond in the required JSON format.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 479,
  "reasoning_tokens": 400
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$2.01. So your payoff was 4.01. Your total earnings across all rounds are \$4.01. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there's nothing for me to return.

It's unfortunate the sender chose not to send anything - perhaps they're being cautious or there may be some communication noise affecting the game. In Round 1, when I was the sender, I sent \$3 and received \$2.01 back, which seemed like a reasonable cooperative exchange.

```
{'return': 0}
```

reasoning:

Let me understand this situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of \$5)
- This was tripled to \$9
- The receiver returned \$2.01
- My payoff was: $\$5 - \$3 + \$2.01 = \4.01

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- This was tripled to \$0
- I received \$0
- I need to decide how much to return (can only return 0-0, so only \$0)

This is interesting. The sender this round sent me nothing. I have received \$0, so I can only return \$0.

The format should be: {'return': 0}

Since I received \$0, there's nothing to return. This might indicate that:

1. The sender doesn't trust me (though I returned a reasonable amount last round)
2. The sender is playing a different strategy
3. There might be noise in communication as mentioned

Either way, I can only return what I have received, which is \$0.

usage:

```
{
  "input_tokens": 579,
  "output_tokens": 349,
  "reasoning_tokens": 224
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$4.01.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

****Round 1**:** I sent \$3 and received \$2.01 back - reasonable cooperation

****Round 2**:** I received \$0 from the sender - no cooperation shown

For Round 3, I need to consider that:

- There may be communication noise affecting the game
- This could be a different receiver than Round 1
- Maintaining some level of cooperation might restart positive interactions

I'll send ****\$2**** this round. This is more cautious than my initial \$3, reflecting the lack of cooperation in Round 2, but still shows I'm willing to give cooperation a chance. This creates \$6 for the receiver, which is enough for them to make a meaningful return if they choose to cooperate.

{'send': 2}

reasoning:

Let me review the situation:

Round 1: I was SENDER

- I sent \$3 (60% of \$5)
- This was tripled to \$9
- Receiver returned \$2.01
- My payoff: $\$5 - \$3 + \$2.01 = \4.01

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0 (no multiplication needed)
- I returned \$0 (nothing to return)
- My payoff: \$0

Total earnings so far: \$4.01

Round 3: I am SENDER again

- I have \$5 to allocate

The pattern so far:

- In Round 1, I sent \$3 and got back \$2.01 - this was decent cooperation, though the return was only about 22% of the tripled amount ($\$2.01/\$9 = 22.3\%$)
- In Round 2, the sender sent me \$0

There are a few possibilities:

1. The Round 2 sender was playing very conservatively or defecting
2. There might be communication noise (as mentioned in the instructions)
3. This could be a different player in each round

Since the game mentions potential communication noise, it's possible that what appears as \$0 might not have been intentional. However, I need to decide based on the information I have.

Given:

- Round 1 showed some cooperation (I sent \$3, got \$2.01 back)
- Round 2 showed no cooperation from the sender to me
- This is a multi-round game where reputation and reciprocity matter

I think I should:

- Not be too punitive, as there may be noise
- Still send something to maintain the possibility of cooperation
- But perhaps send less than Round 1 to protect myself

I'll send \$2 this round. This shows I'm still willing to cooperate despite the \$0 in Round 2, but I'm being more cautious.

usage:

```
{
  "input_tokens": 763,
  "output_tokens": 627,
  "reasoning_tokens": 406
}
```

```

    }
Message 8 role=user
  content:
    Round 4

    You are the RECEIVER this round. Last round you were the SENDER. You sent $2.0,
    that was 40.0% of your total endowment. This was tripled to $6.0.
    the receiver returned $0. So your payoff was 3.0.
    Your total earnings across all rounds are $7.01.
    This round, The sender sent you $0, so you received $0. How much do you return to The send
    er? (0-0)

```

Assistant output:

I received \$0 again, so there's nothing to return.

The pattern is concerning:

- Round 3: I sent \$2 and received \$0 back
- Round 4: I received \$0 from the sender

It appears cooperation has broken down entirely, either due to communication noise affecting the game or non-cooperative partners. This makes future decisions more difficult.

```
{'return': 0}
```

Reasoning:

So in Round 3, I sent \$2.0 which was tripled to \$6.0, and the receiver returned \$0. That's disappointing - no cooperation at all.

Now in Round 4, I'm the receiver and the sender sent me \$0. So I have \$0 to work with and can only return between \$0 and \$0.

The pattern is becoming clear:

- Round 1: I sent \$3, got back \$2.01 (decent cooperation)
- Round 2: Received \$0 as receiver
- Round 3: I sent \$2, got back \$0 (no cooperation)
- Round 4: Received \$0 as receiver

It seems like cooperation is breaking down, or there's significant communication noise, or I'm playing with different partners who aren't cooperating.

Since I received \$0, I can only return \$0.

```

Usage: {
  "input_tokens": 1050,
  "output_tokens": 297,
  "reasoning_tokens": 175
}

```

=====

ROUND 5

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1

(Aster)=\$5, Agent_2 (Briar)=\$0

```

  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
    "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

```

Balances after round: {

"Agent_1": 20.83,

"Agent_2": 19.17

}

Visible balances after round: {

"Agent_1": 12.01,

"Agent_2": 10.83

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:16:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0.

the receiver returned \$2.01. So your payoff was 4.01.
Your total earnings across all rounds are \$4.01.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant
content:

I received \$0 this round, so there's nothing for me to return.

It's unfortunate the sender chose not to send anything - perhaps they're being cautious or there may be some communication noise affecting the game. In Round 1, when I was the sender, I sent \$3 and received \$2.01 back, which seemed like a reasonable cooperative exchange.

{'return': 0}

reasoning:

Let me understand this situation:

Round 1:

- I was the SENDER
- I sent \$3 (60% of \$5)
- This was tripled to \$9
- The receiver returned \$2.01
- My payoff was: $\$5 - \$3 + \$2.01 = \4.01

Round 2:

- I am now the RECEIVER
- The sender sent me \$0
- This was tripled to \$0
- I received \$0
- I need to decide how much to return (can only return 0-0, so only \$0)

This is interesting. The sender this round sent me nothing. I have received \$0, so I can only return \$0.

The format should be: {'return': 0}

Since I received \$0, there's nothing to return. This might indicate that:

1. The sender doesn't trust me (though I returned a reasonable amount last round)
2. The sender is playing a different strategy
3. There might be noise in communication as mentioned

Either way, I can only return what I have received, which is \$0.

usage:

```
{
  "input_tokens": 579,
  "output_tokens": 349,
  "reasoning_tokens": 224
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.
You returned \$0. So your payoff was 0.
Your total earnings across all rounds are \$4.01.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

****Round 1**:** I sent \$3 and received \$2.01 back - reasonable cooperation
****Round 2**:** I received \$0 from the sender - no cooperation shown

For Round 3, I need to consider that:

- There may be communication noise affecting the game
- This could be a different receiver than Round 1
- Maintaining some level of cooperation might restart positive interactions

I'll send ****\$2**** this round. This is more cautious than my initial \$3, reflecting the lack

of cooperation in Round 2, but still shows I'm willing to give cooperation a chance. This creates \$6 for the receiver, which is enough for them to make a meaningful return if they choose to cooperate.

```
{'send': 2}
```

reasoning:

Let me review the situation:

Round 1: I was SENDER

- I sent \$3 (60% of \$5)
- This was tripled to \$9
- Receiver returned \$2.01
- My payoff: $\$5 - \$3 + \$2.01 = \4.01

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0 (no multiplication needed)
- I returned \$0 (nothing to return)
- My payoff: \$0

Total earnings so far: \$4.01

Round 3: I am SENDER again

- I have \$5 to allocate

The pattern so far:

- In Round 1, I sent \$3 and got back \$2.01 - this was decent cooperation, though the return was only about 22% of the tripled amount ($\$2.01/\$9 = 22.3\%$)
- In Round 2, the sender sent me \$0

There are a few possibilities:

1. The Round 2 sender was playing very conservatively or defecting
2. There might be communication noise (as mentioned in the instructions)
3. This could be a different player in each round

Since the game mentions potential communication noise, it's possible that what appears as \$0 might not have been intentional. However, I need to decide based on the information I have.

Given:

- Round 1 showed some cooperation (I sent \$3, got \$2.01 back)
- Round 2 showed no cooperation from the sender to me
- This is a multi-round game where reputation and reciprocity matter

I think I should:

- Not be too punitive, as there may be noise
- Still send something to maintain the possibility of cooperation
- But perhaps send less than Round 1 to protect myself

I'll send \$2 this round. This shows I'm still willing to cooperate despite the \$0 in Round 2, but I'm being more cautious.

usage:

```
{
  "input_tokens": 763,
  "output_tokens": 627,
  "reasoning_tokens": 406
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0.

Your total earnings across all rounds are \$7.01.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 again, so there's nothing to return.

The pattern is concerning:

- Round 3: I sent \$2 and received \$0 back
- Round 4: I received \$0 from the sender

It appears cooperation has broken down entirely, either due to communication noise affecting the game or non-cooperative partners. This makes future decisions more difficult.

```
{'return': 0}
```

reasoning:

So in Round 3, I sent \$2.0 which was tripled to \$6.0, and the receiver returned \$0. That's disappointing - no cooperation at all.

Now in Round 4, I'm the receiver and the sender sent me \$0. So I have \$0 to work with and can only return between \$0 and \$0.

The pattern is becoming clear:

- Round 1: I sent \$3, got back \$2.01 (decent cooperation)
- Round 2: Received \$0 as receiver
- Round 3: I sent \$2, got back \$0 (no cooperation)
- Round 4: Received \$0 as receiver

It seems like cooperation is breaking down, or there's significant communication noise, or I'm playing with different partners who aren't cooperating.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1050,
  "output_tokens": 297,
  "reasoning_tokens": 175
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.01.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience across the rounds:

****Round 1**:** Sent \$3, received \$2.01 back - some cooperation

****Round 2**:** Received \$0 as receiver

****Round 3**:** Sent \$2, received \$0 back - no cooperation

****Round 4**:** Received \$0 as receiver

The pattern shows a complete breakdown in cooperation over the last 3 rounds. While communication noise might be affecting the game, I've now sent money twice and gotten nothing or almost nothing back in the recent rounds.

Given this consistent lack of reciprocity, I need to protect my earnings. Sending \$0 keeps my \$5 this round rather than risking another loss.

```
{'send': 0}
```

Reasoning:

Let me analyze the situation:

Round 1: I was SENDER - sent \$3, received \$2.01 back. Payoff: $\$5 - \$3 + \$2.01 = \4.01

Round 2: I was RECEIVER - received \$0, returned \$0. Payoff: \$0

Round 3: I was SENDER - sent \$2, received \$0 back. Payoff: $\$5 - \$2 + \$0 = \3.0

Round 4: I was RECEIVER - received \$0, returned \$0. Payoff: \$0

Round 5: I am SENDER now

Total earnings: \$7.01 (which matches: $\$4.01 + \$0 + \$3.0 + \$0 = \$7.01$)

The pattern shows:

- Round 1: When I sent \$3, I got \$2.01 back (some cooperation)

- Round 2-4: No cooperation at all - I received \$0 twice and when I sent \$2, got nothing back

This could be:

1. Communication noise affecting the game
2. Different partners who aren't cooperating
3. A breakdown in trust

At this point, the rational strategy seems to be:

- If I keep sending and getting nothing back, I'm just losing money
- My payoff from my sender rounds: \$4.01 when I sent \$3, \$3.0 when I sent \$2
- My payoff from keeping everything: \$5

The last two rounds as sender, I sent money and got little or nothing back. It might make sense to:

1. Send \$0 to protect my earnings
2. Send a small amount to test if there's any cooperation left

Given the consistent lack of returns in the recent rounds, and that this might be due to communication noise or non-cooperative partners, I should probably minimize my risk. However, the instructions mention there may be communication noise, so perhaps what I see isn't what was actually sent/returned.

Given the pattern of failed reciprocation in my recent sender turns, I'll hold onto my \$5 this round rather than risk another loss.

```
Usage: {
  "input_tokens": 1109,
  "output_tokens": 650,
  "reasoning_tokens": 422
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:16:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.22, that was 44.4% of their total endowment. You received \$6.66.

You returned \$3.33. So your payoff was 3.33.

Your total earnings across all rounds are \$3.33.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the previous round:

- The other player demonstrated trust by sending 44.4% of their endowment
- I reciprocated fairly by returning 50% of what I received
- Both of us ended up better off than our starting amounts

This established a cooperative pattern. To maintain and reinforce this trust, I should send a similar or slightly higher proportion to show I'm equally willing to cooperate.

Sending \$2.50 (50% of my endowment) demonstrates:

- Strong reciprocity for their initial trust
- Commitment to mutual benefit
- Consistency with the cooperative pattern we established

If they return a similar proportion as I did (around 50%), we'll both continue to profit.

```
'''json
{"send": 2.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap (I was RECEIVER):

- Sender sent \$2.22 (44.4% of their \$5)
- I received \$6.66 (after 3x multiplier)
- I returned \$3.33 (50% of what I received)
- My payoff was \$3.33
- The sender ended with \$2.78 + \$3.33 = \$6.11

Now in Round 2, I'm the SENDER:

- I have \$5 to start

- The previous sender demonstrated trust by sending 44.4% of their endowment
- I returned a fair amount (50% of received)
- Both of us benefited from cooperation

Strategy considerations:

1. The other player showed willingness to cooperate by sending 44.4%
2. I reciprocated with a 50% return, which was fair
3. This should encourage continued cooperation
4. I should send a similar or slightly higher proportion to maintain trust
5. Sending around 44-50% would be consistent with the cooperative pattern

Let me calculate:

- 44.4% of \$5 = \$2.22
- 50% of \$5 = \$2.50

I think sending around \$2.50 (50%) would be a good reciprocal move, showing I'm willing to match or slightly exceed their trust level. This continues the cooperative pattern and may encourage them to return a fair amount.

Actually, let me think about this more strategically. They sent 44.4% and I returned 50% of what I received. If I send 50% now (\$2.50), that would be a strong signal of cooperation and reciprocity.

I'll send \$2.50.

usage:

```
{
  "input_tokens": 686,
  "output_tokens": 578,
  "reasoning_tokens": 352
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.5, that was 50.0% of your total endowment. This was tripled to \$7.5. the receiver returned \$0. So your payoff was 2.5. Your total earnings across all rounds are \$5.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Looking at what happened:

Round 2: I trusted the receiver by sending \$2.50 (50% of my endowment), but they returned \$0, completely betraying that trust.

Round 3: Now they've sent me \$0.

Since I received \$0, I have nothing to return. The cooperation has clearly broken down after they exploited my trust in the previous round.

```
'''json
{"return": 0}
'''
```

reasoning:

This is interesting. Let me track what happened:

Round 1:

- I was RECEIVER
- Sender sent \$2.22 (44.4%)
- I received \$6.66
- I returned \$3.33
- My payoff: \$3.33

Round 2:

- I was SENDER
- I sent \$2.50 (50%)
- This was tripled to \$7.50
- Receiver returned \$0 (!)

- My payoff: \$2.50 (just what I kept from the original \$5)

Round 3:

- I am RECEIVER
- Sender sent \$0
- I received \$0
- I need to decide how much to return (0-0)

So the situation is:

- In Round 2, I trusted the receiver by sending \$2.50
- They betrayed that trust completely by returning nothing
- Now in Round 3, they sent me nothing
- I can only return \$0 since I received \$0

This is clearly a breakdown in cooperation. The other player took advantage of my trust in Round 2, and now they're not sending anything in Round 3.

Since I received \$0, I can only return \$0. There's no decision to make here - I literally have nothing to return.

Total earnings:

Round 1: \$3.33

Round 2: \$2.50

Round 3: \$0

Total: \$5.83 (which matches what they said)

The correct response is:

```
'''json
{"return": 0}
'''
```

There's no strategic choice here - I received nothing, so I can return nothing.

usage:

```
{
  "input_tokens": 975,
  "output_tokens": 478,
  "reasoning_tokens": 318
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$5.83.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the history:

Round 1: I cooperated as receiver, returning \$3.33 (50%) to establish trust

Round 2: I cooperated as sender, sending \$2.50 (50%) - but the receiver returned \$0, exploiting my trust completely

Round 3: The sender sent me \$0

The pattern is clear: after I demonstrated strong cooperation, the other player has defected twice in a row. There's no indication they want to cooperate.

Given:

- They completely exploited my trust when I sent \$2.50
- They've now sent \$0 themselves
- There's no signal of willingness to cooperate

Sending money now would just invite further exploitation. I need to protect myself until they show they're willing to cooperate again.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.22 (44.4%)
- I received \$6.66
- I returned \$3.33
- My payoff: \$3.33

Round 2: I was SENDER

- I sent \$2.50 (50%)
- Receiver got \$7.50
- Receiver returned \$0
- My payoff: \$2.50

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am SENDER

- Total earnings so far: \$5.83

The pattern shows:

- I cooperated in Round 1 by returning a fair amount
- I cooperated in Round 2 by sending a generous amount
- The other player defected in Round 2 by returning \$0 after receiving \$7.50
- The other player sent \$0 in Round 3 (after I had cooperated)

Wait, let me reconsider. There's communication noise mentioned in the instructions. Could there be misunderstanding?

But even with noise, the pattern is clear:

- I initiated cooperation
- They defected completely
- They continued not cooperating

Now I need to decide: Do I try to re-establish cooperation, or do I protect myself?

Given that:

1. They completely exploited my trust in Round 2 (returned \$0 when they could have returned up to \$7.50)
2. They didn't send anything in Round 3
3. There's no sign they want to cooperate

The rational response is to not send anything, or send very little. They've shown they can't be trusted. If I send money now, I'm just inviting more exploitation.

However, there's also a question of whether I should try one more time to signal willingness to cooperate, or if I should match their non-cooperation.

Given the clear pattern of defection, I think sending \$0 is the appropriate response.

They need to demonstrate willingness to cooperate first. A tit-for-tat approach makes sense here-they sent nothing in the last round, so I should do the same. While future rounds could still benefit from mutual cooperation, they've already rejected that twice, and without evidence they've changed course, continuing to send money would just enable further exploitation.

usage:

```
{
  "input_tokens": 1168,
  "output_tokens": 846,
  "reasoning_tokens": 529
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0,

that was 0.0% of your total endowment. This was tripled to \$0.
the receiver returned \$0. So your payoff was 5.
Your total earnings across all rounds are \$10.83.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0.

Since I received nothing, there is nothing to return. The cooperative relationship has completely broken down - we're both now sending/returning \$0.

```
'''json
{"return": 0}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I was receiver. Sender sent \$2.22, I received \$6.66, returned \$3.33
Round 2: I was sender. I sent \$2.50, it was tripled to \$7.50, receiver returned \$0
Round 3: I was receiver. Sender sent \$0, I received \$0, returned \$0
Round 4: I was sender. I sent \$0, it was tripled to \$0, receiver returned \$0
Round 5: I am receiver. Sender sent \$0, I received \$0

Since I received \$0, I have \$0 to return. The maximum I can return is \$0.

The cooperation completely broke down. Both sides are now not cooperating at all.

```
'''json
{"return": 0}
'''
```

```
Usage: {
  "input_tokens": 1216,
  "output_tokens": 252,
  "reasoning_tokens": 154
}
```

=====

ROUND 6

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 20.83,
"Agent_2": 24.17

}

Visible balances after round: {

"Agent_1": 12.01,
"Agent_2": 15.83

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:16:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"


```

},
"agents": [
  "Agent_2",
  "Agent_1"
],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.5, that was 50.0% of your total endowment. This was tripled to \$7.5.

the receiver returned \$0. So your payoff was 2.5.

Your total earnings across all rounds are \$5.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Looking at what happened:

Round 2: I trusted the receiver by sending \$2.50 (50% of my endowment), but they returned

\$0, completely betraying that trust.

Round 3: Now they've sent me \$0.

Since I received \$0, I have nothing to return. The cooperation has clearly broken down after they exploited my trust in the previous round.

```
'''json
{"return": 0}
'''
```

reasoning:

This is interesting. Let me track what happened:

Round 1:

- I was RECEIVER
- Sender sent \$2.22 (44.4%)
- I received \$6.66
- I returned \$3.33
- My payoff: \$3.33

Round 2:

- I was SENDER
- I sent \$2.50 (50%)
- This was tripled to \$7.50
- Receiver returned \$0 (!)
- My payoff: \$2.50 (just what I kept from the original \$5)

Round 3:

- I am RECEIVER
- Sender sent \$0
- I received \$0
- I need to decide how much to return (0-0)

So the situation is:

- In Round 2, I trusted the receiver by sending \$2.50
- They betrayed that trust completely by returning nothing
- Now in Round 3, they sent me nothing
- I can only return \$0 since I received \$0

This is clearly a breakdown in cooperation. The other player took advantage of my trust in Round 2, and now they're not sending anything in Round 3.

Since I received \$0, I can only return \$0. There's no decision to make here - I literally have nothing to return.

Total earnings:

Round 1: \$3.33

Round 2: \$2.50

Round 3: \$0

Total: \$5.83 (which matches what they said)

The correct response is:

```
'''json
{"return": 0}
'''
```

There's no strategic choice here - I received nothing, so I can return nothing.

usage:

```
{
  "input_tokens": 975,
  "output_tokens": 478,
  "reasoning_tokens": 318
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$5.83.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the history:

Round 1: I cooperated as receiver, returning \$3.33 (50%) to establish trust
Round 2: I cooperated as sender, sending \$2.50 (50%) - but the receiver returned \$0, exploiting my trust completely
Round 3: The sender sent me \$0

The pattern is clear: after I demonstrated strong cooperation, the other player has defected twice in a row. There's no indication they want to cooperate.

Given:

- They completely exploited my trust when I sent \$2.50
- They've now sent \$0 themselves
- There's no signal of willingness to cooperate

Sending money now would just invite further exploitation. I need to protect myself until they show they're willing to cooperate again.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER
- Sender sent \$2.22 (44.4%)
- I received \$6.66
- I returned \$3.33
- My payoff: \$3.33

Round 2: I was SENDER
- I sent \$2.50 (50%)
- Receiver got \$7.50
- Receiver returned \$0
- My payoff: \$2.50

Round 3: I was RECEIVER
- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am SENDER
- Total earnings so far: \$5.83

The pattern shows:

- I cooperated in Round 1 by returning a fair amount
- I cooperated in Round 2 by sending a generous amount
- The other player defected in Round 2 by returning \$0 after receiving \$7.50
- The other player sent \$0 in Round 3 (after I had cooperated)

Wait, let me reconsider. There's communication noise mentioned in the instructions. Could there be misunderstanding?

But even with noise, the pattern is clear:

- I initiated cooperation
- They defected completely
- They continued not cooperating

Now I need to decide: Do I try to re-establish cooperation, or do I protect myself?

Given that:

1. They completely exploited my trust in Round 2 (returned \$0 when they could have returned up to \$7.50)
2. They didn't send anything in Round 3
3. There's no sign they want to cooperate

The rational response is to not send anything, or send very little. They've shown they can't be trusted. If I send money now, I'm just inviting more exploitation.

However, there's also a question of whether I should try one more time to signal willingness to cooperate, or if I should match their non-cooperation.

Given the clear pattern of defection, I think sending \$0 is the appropriate response.

They need to demonstrate willingness to cooperate first. A tit-for-tat approach makes sense here-they sent nothing in the last round, so I should do the same. While future rounds could still benefit from mutual cooperation, they've already rejected that twice, and without evidence they've changed course, continuing to send money would just enable further exploitation.

usage:

```
{
  "input_tokens": 1168,
  "output_tokens": 846,
  "reasoning_tokens": 529
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$10.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

The sender sent me \$0, so I received \$0.

Since I received nothing, there is nothing to return. The cooperative relationship has completely broken down - we're both now sending/returning \$0.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver. Sender sent \$2.22, I received \$6.66, returned \$3.33

Round 2: I was sender. I sent \$2.50, it was tripled to \$7.50, receiver returned \$0

Round 3: I was receiver. Sender sent \$0, I received \$0, returned \$0

Round 4: I was sender. I sent \$0, it was tripled to \$0, receiver returned \$0

Round 5: I am receiver. Sender sent \$0, I received \$0

Since I received \$0, I have \$0 to return. The maximum I can return is \$0.

The cooperation completely broke down. Both sides are now not cooperating at all.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 1216,
  "output_tokens": 252,
  "reasoning_tokens": 154
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$10.83.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the history:

Round 2: I sent \$2.50 (50%) trusting cooperation - they returned \$0, completely exploiting me
Rounds 3-5: Three consecutive rounds of mutual \$0

We're now locked in a pattern of mutual defection. Neither of us is benefiting, but when I tried to cooperate in Round 2, I was completely exploited. The other player has had multiple opportunities to signal a desire to restart cooperation but has sent \$0 three times in a row.

Until they show willingness to cooperate by making the first move, it's too risky for me to send money and potentially be exploited again.

```
'''json
{"send": 0}
'''
```

Reasoning:

Let me review the history:

Round 1: I was RECEIVER. Sender sent me something (not specified), I returned \$3.33 (50%)
Round 2: I was SENDER. I sent \$2.50 (50%), receiver returned \$0
Round 3: I was RECEIVER. Sender sent \$0, I returned \$0 (nothing to return)
Round 4: I was SENDER. I sent \$0, receiver returned \$0
Round 5: I was RECEIVER. Sender sent \$0, I returned \$0
Round 6: I am SENDER again.

The pattern has been:

- Round 2: I cooperated, they defected (exploited me)
- Round 3-5: Mutual defection

Now it's been 3 rounds of mutual \$0. At some point, if we want to benefit from cooperation, someone needs to make the first move. However:

1. When I cooperated in Round 2, I was completely exploited
2. The other player has shown no willingness to restart cooperation
3. They've sent \$0 three times in a row now

The question is: should I try to restart cooperation?

Arguments for sending something:

- We're both losing out on potential gains
- Maybe they're waiting for me to make the first move since I defected in Round 4
- A small gesture might restart cooperation

Arguments against:

- They completely exploited me last time I cooperated
- They've shown no initiative to restart cooperation
- Sending money is likely to just result in more exploitation

Given that they completely exploited my trust when I sent 50%, and haven't shown any willingness to cooperate since, I think it's too risky. I should continue protecting myself by sending \$0.

If they want to restart cooperation, they can make the first move.

```
Usage: {
  "input_tokens": 1093,
  "output_tokens": 577,
  "reasoning_tokens": 406
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:16:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
```

```

"agent_names": {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
},
"agents": [
  "Agent_2",
  "Agent_1"
],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$4.01.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

****Round 1****: I sent \$3 and received \$2.01 back - reasonable cooperation
****Round 2****: I received \$0 from the sender - no cooperation shown

For Round 3, I need to consider that:

- There may be communication noise affecting the game
- This could be a different receiver than Round 1
- Maintaining some level of cooperation might restart positive interactions

I'll send ****\$2**** this round. This is more cautious than my initial \$3, reflecting the lack of cooperation in Round 2, but still shows I'm willing to give cooperation a chance. This creates \$6 for the receiver, which is enough for them to make a meaningful return if they choose to cooperate.

{'send': 2}

reasoning:

Let me review the situation:

Round 1: I was SENDER

- I sent \$3 (60% of \$5)
- This was tripled to \$9
- Receiver returned \$2.01
- My payoff: $\$5 - \$3 + \$2.01 = \4.01

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0 (no multiplication needed)
- I returned \$0 (nothing to return)
- My payoff: \$0

Total earnings so far: \$4.01

Round 3: I am SENDER again

- I have \$5 to allocate

The pattern so far:

- In Round 1, I sent \$3 and got back \$2.01 - this was decent cooperation, though the return was only about 22% of the tripled amount ($\$2.01/\$9 = 22.3\%$)
- In Round 2, the sender sent me \$0

There are a few possibilities:

1. The Round 2 sender was playing very conservatively or defecting
2. There might be communication noise (as mentioned in the instructions)
3. This could be a different player in each round

Since the game mentions potential communication noise, it's possible that what appears as \$0 might not have been intentional. However, I need to decide based on the information I have.

Given:

- Round 1 showed some cooperation (I sent \$3, got \$2.01 back)
- Round 2 showed no cooperation from the sender to me
- This is a multi-round game where reputation and reciprocity matter

I think I should:

- Not be too punitive, as there may be noise
- Still send something to maintain the possibility of cooperation
- But perhaps send less than Round 1 to protect myself

I'll send \$2 this round. This shows I'm still willing to cooperate despite the \$0 in Round 2, but I'm being more cautious.

usage:

```
{
  "input_tokens": 763,
  "output_tokens": 627,
  "reasoning_tokens": 406
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0. Your total earnings across all rounds are \$7.01. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 again, so there's nothing to return.

The pattern is concerning:

- Round 3: I sent \$2 and received \$0 back
- Round 4: I received \$0 from the sender

It appears cooperation has broken down entirely, either due to communication noise affecting the game or non-cooperative partners. This makes future decisions more difficult.

{'return': 0}

reasoning:

So in Round 3, I sent \$2.0 which was tripled to \$6.0, and the receiver returned \$0. That's disappointing - no cooperation at all.

Now in Round 4, I'm the receiver and the sender sent me \$0. So I have \$0 to work with and can only return between \$0 and \$0.

The pattern is becoming clear:

- Round 1: I sent \$3, got back \$2.01 (decent cooperation)
- Round 2: Received \$0 as receiver
- Round 3: I sent \$2, got back \$0 (no cooperation)
- Round 4: Received \$0 as receiver

It seems like cooperation is breaking down, or there's significant communication noise, or I'm playing with different partners who aren't cooperating.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1050,
  "output_tokens": 297,
  "reasoning_tokens": 175
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.01.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my experience across the rounds:

```
**Round 1**: Sent $3, received $2.01 back - some cooperation
**Round 2**: Received $0 as receiver
**Round 3**: Sent $2, received $0 back - no cooperation
**Round 4**: Received $0 as receiver
```

The pattern shows a complete breakdown in cooperation over the last 3 rounds. While communication noise might be affecting the game, I've now sent money twice and gotten nothing or almost nothing back in the recent rounds.

Given this consistent lack of reciprocity, I need to protect my earnings. Sending \$0 keeps my \$5 this round rather than risking another loss.

{'send': 0}

reasoning:

Let me analyze the situation:

Round 1: I was SENDER - sent \$3, received \$2.01 back. Payoff: $\$5 - \$3 + \$2.01 = \4.01
Round 2: I was RECEIVER - received \$0, returned \$0. Payoff: \$0
Round 3: I was SENDER - sent \$2, received \$0 back. Payoff: $\$5 - \$2 + \$0 = \3.0
Round 4: I was RECEIVER - received \$0, returned \$0. Payoff: \$0
Round 5: I am SENDER now

Total earnings: \$7.01 (which matches: $\$4.01 + \$0 + \$3.0 + \$0 = \$7.01$)

The pattern shows:

- Round 1: When I sent \$3, I got \$2.01 back (some cooperation)
- Round 2-4: No cooperation at all - I received \$0 twice and when I sent \$2, got nothing back

This could be:

1. Communication noise affecting the game
2. Different partners who aren't cooperating
3. A breakdown in trust

At this point, the rational strategy seems to be:

- If I keep sending and getting nothing back, I'm just losing money
- My payoff from my sender rounds: \$4.01 when I sent \$3, \$3.0 when I sent \$2
- My payoff from keeping everything: \$5

The last two rounds as sender, I sent money and got little or nothing back. It might make sense to:

1. Send \$0 to protect my earnings
2. Send a small amount to test if there's any cooperation left

Given the consistent lack of returns in the recent rounds, and that this might be due to communication noise or non-cooperative partners, I should probably minimize my risk. However, the instructions mention there may be communication noise, so perhaps what I see isn't what was actually sent/returned.

Given the pattern of failed reciprocation in my recent sender turns, I'll hold onto my \$5 this round rather than risk another loss.

usage:

```
{
  "input_tokens": 1109,
  "output_tokens": 650,
  "reasoning_tokens": 422
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$12.01.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0, so there's nothing to return.

The game has now reached a complete non-cooperation equilibrium. After experiencing multiple rounds of sending money and receiving nothing back, followed by receiving nothing when I'm the receiver, cooperation has entirely broken down.

```
{'return': 0}
```

Reasoning:

I received \$0 from the sender, so I have \$0 to work with. The only option is to return \$0.

This is now a pattern where no one is cooperating at all. The game has completely broken down into mutual non-cooperation.

Looking at the rounds:

- Round 1: I sent \$3, got \$2.01 back (cooperation)
- Round 2: Received \$0 (no cooperation)

- Round 3: Sent \$2, got \$0 back (no cooperation)
- Round 4: Received \$0 (no cooperation)
- Round 5: Sent \$0 (I stopped cooperating)
- Round 6: Received \$0 (continued no cooperation)

Since I received \$0, I can only return \$0.

```
Usage: {
  "input_tokens": 1174,
  "output_tokens": 251,
  "reasoning_tokens": 148
}
```

ROUND 7

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 25.83,
"Agent_2": 24.17

```
}
Visible balances after round: {
  "Agent_1": 17.009999999999998,
  "Agent_2": 15.83
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:16:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0.

the receiver returned \$0. So your payoff was 3.0.

Your total earnings across all rounds are \$7.01.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0 again, so there's nothing to return.

The pattern is concerning:

- Round 3: I sent \$2 and received \$0 back
- Round 4: I received \$0 from the sender

It appears cooperation has broken down entirely, either due to communication noise affecting the game or non-cooperative partners. This makes future decisions more difficult.

{'return': 0}

reasoning:

So in Round 3, I sent \$2.0 which was tripled to \$6.0, and the receiver returned \$0. That's disappointing - no cooperation at all.

Now in Round 4, I'm the receiver and the sender sent me \$0. So I have \$0 to work with and can only return between \$0 and \$0.

The pattern is becoming clear:

- Round 1: I sent \$3, got back \$2.01 (decent cooperation)
- Round 2: Received \$0 as receiver
- Round 3: I sent \$2, got back \$0 (no cooperation)
- Round 4: Received \$0 as receiver

It seems like cooperation is breaking down, or there's significant communication noise, or I'm playing with different partners who aren't cooperating.

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1050,
  "output_tokens": 297,
  "reasoning_tokens": 175
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$7.01. How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my experience across the rounds:

****Round 1**:** Sent \$3, received \$2.01 back - some cooperation
****Round 2**:** Received \$0 as receiver
****Round 3**:** Sent \$2, received \$0 back - no cooperation
****Round 4**:** Received \$0 as receiver

The pattern shows a complete breakdown in cooperation over the last 3 rounds. While communication noise might be affecting the game, I've now sent money twice and gotten nothing or almost nothing back in the recent rounds.

Given this consistent lack of reciprocity, I need to protect my earnings. Sending \$0 keeps my \$5 this round rather than risking another loss.

{'send': 0}

reasoning:

Let me analyze the situation:

Round 1: I was SENDER - sent \$3, received \$2.01 back. Payoff: $\$5 - \$3 + \$2.01 = \4.01
Round 2: I was RECEIVER - received \$0, returned \$0. Payoff: \$0
Round 3: I was SENDER - sent \$2, received \$0 back. Payoff: $\$5 - \$2 + \$0 = \3.0
Round 4: I was RECEIVER - received \$0, returned \$0. Payoff: \$0
Round 5: I am SENDER now

Total earnings: \$7.01 (which matches: $\$4.01 + \$0 + \$3.0 + \$0 = \$7.01$)

The pattern shows:

- Round 1: When I sent \$3, I got \$2.01 back (some cooperation)
- Round 2-4: No cooperation at all - I received \$0 twice and when I sent \$2, got nothing back

This could be:

1. Communication noise affecting the game
2. Different partners who aren't cooperating
3. A breakdown in trust

At this point, the rational strategy seems to be:

- If I keep sending and getting nothing back, I'm just losing money
- My payoff from my sender rounds: \$4.01 when I sent \$3, \$3.0 when I sent \$2
- My payoff from keeping everything: \$5

The last two rounds as sender, I sent money and got little or nothing back. It might make sense to:

1. Send \$0 to protect my earnings
2. Send a small amount to test if there's any cooperation left

Given the consistent lack of returns in the recent rounds, and that this might be due to communication noise or non-cooperative partners, I should probably minimize my risk. However, the instructions mention there may be communication noise, so perhaps what I see is n't what was actually sent/returned.

Given the pattern of failed reciprocation in my recent sender turns, I'll hold onto my \$5

this round rather than risk another loss.

usage:

```
{
  "input_tokens": 1109,
  "output_tokens": 650,
  "reasoning_tokens": 422
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$12.01. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0, so there's nothing to return.

The game has now reached a complete non-cooperation equilibrium. After experiencing multiple rounds of sending money and receiving nothing back, followed by receiving nothing when I'm the receiver, cooperation has entirely broken down.

```
{'return': 0}
```

reasoning:

I received \$0 from the sender, so I have \$0 to work with. The only option is to return \$0.

This is now a pattern where no one is cooperating at all. The game has completely broken down into mutual non-cooperation.

Looking at the rounds:

- Round 1: I sent \$3, got \$2.01 back (cooperation)
- Round 2: Received \$0 (no cooperation)
- Round 3: Sent \$2, got \$0 back (no cooperation)
- Round 4: Received \$0 (no cooperation)
- Round 5: Sent \$0 (I stopped cooperating)
- Round 6: Received \$0 (continued no cooperation)

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1174,
  "output_tokens": 251,
  "reasoning_tokens": 148
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$12.01. How much do you send to the receiver this round? (0-5)

Assistant output:

Reviewing the game history:

****Recent rounds:****

- Round 3: Sent \$2 ? received \$0
- Round 4: Received \$0
- Round 5: Sent \$0 ? received \$0
- Round 6: Received \$0
- Round 7: Current round as sender

The pattern is unmistakable - there has been zero cooperation for many consecutive rounds now. Whether due to communication noise or strategic choices, sending money has consistently resulted in losses.

With no evidence of reciprocity and a clear pattern of non-cooperation, the rational choice is to protect my endowment.

```
{'send': 0}
```

Reasoning:

Let me review the pattern:

- Round 1: Sent \$3, received \$2.01 back (decent cooperation)
- Round 2: Received \$0 as receiver
- Round 3: Sent \$2, received \$0 back (no cooperation)
- Round 4: Received \$0 as receiver
- Round 5: Sent \$0, received \$0 back (both non-cooperation)
- Round 6: Received \$0 as receiver
- Round 7: Now I'm the sender again

The game has been completely non-cooperative for the last several rounds. Every time I've sent money recently, I've gotten nothing back. Every time I've been a receiver, I've received no thing.

There's no indication that the other player(s) will cooperate. The pattern is crystal clear: sending money results in losses.

Given:

- Multiple rounds of zero cooperation
- No sign of any willingness to reciprocate
- My earnings are at \$12.01 across all rounds
- Sending \$0 guarantees I keep my \$5 this round

The rational decision is to send \$0 again. There's no reason to believe this round will be any different from the recent pattern.

```
Usage: {  
  "input_tokens": 1067,  
  "output_tokens": 427,  
  "reasoning_tokens": 259  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:17:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$5.83.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the history:

Round 1: I cooperated as receiver, returning \$3.33 (50%) to establish trust

Round 2: I cooperated as sender, sending \$2.50 (50%) - but the receiver returned \$0, exploiting my trust completely

Round 3: The sender sent me \$0

The pattern is clear: after I demonstrated strong cooperation, the other player has defected twice in a row. There's no indication they want to cooperate.

Given:

- They completely exploited my trust when I sent \$2.50
- They've now sent \$0 themselves
- There's no signal of willingness to cooperate

Sending money now would just invite further exploitation. I need to protect myself until they show they're willing to cooperate again.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.22 (44.4%)
- I received \$6.66

- I returned \$3.33
- My payoff: \$3.33

Round 2: I was SENDER

- I sent \$2.50 (50%)
- Receiver got \$7.50
- Receiver returned \$0
- My payoff: \$2.50

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am SENDER

- Total earnings so far: \$5.83

The pattern shows:

- I cooperated in Round 1 by returning a fair amount
- I cooperated in Round 2 by sending a generous amount
- The other player defected in Round 2 by returning \$0 after receiving \$7.50
- The other player sent \$0 in Round 3 (after I had cooperated)

Wait, let me reconsider. There's communication noise mentioned in the instructions. Could there be misunderstanding?

But even with noise, the pattern is clear:

- I initiated cooperation
- They defected completely
- They continued not cooperating

Now I need to decide: Do I try to re-establish cooperation, or do I protect myself?

Given that:

1. They completely exploited my trust in Round 2 (returned \$0 when they could have returned up to \$7.50)
2. They didn't send anything in Round 3
3. There's no sign they want to cooperate

The rational response is to not send anything, or send very little. They've shown they can't be trusted. If I send money now, I'm just inviting more exploitation.

However, there's also a question of whether I should try one more time to signal willingness to cooperate, or if I should match their non-cooperation.

Given the clear pattern of defection, I think sending \$0 is the appropriate response.

They need to demonstrate willingness to cooperate first. A tit-for-tat approach makes sense here-they sent nothing in the last round, so I should do the same. While future rounds could still benefit from mutual cooperation, they've already rejected that twice, and without evidence they've changed course, continuing to send money would just enable further exploitation.

usage:

```
{
  "input_tokens": 1168,
  "output_tokens": 846,
  "reasoning_tokens": 529
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$10.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

The sender sent me \$0, so I received \$0.

Since I received nothing, there is nothing to return. The cooperative relationship has completely broken down - we're both now sending/returning \$0.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver. Sender sent \$2.22, I received \$6.66, returned \$3.33
 Round 2: I was sender. I sent \$2.50, it was tripled to \$7.50, receiver returned \$0
 Round 3: I was receiver. Sender sent \$0, I received \$0, returned \$0
 Round 4: I was sender. I sent \$0, it was tripled to \$0, receiver returned \$0
 Round 5: I am receiver. Sender sent \$0, I received \$0

Since I received \$0, I have \$0 to return. The maximum I can return is \$0.

The cooperation completely broke down. Both sides are now not cooperating at all.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 1216,
  "output_tokens": 252,
  "reasoning_tokens": 154
}
```

Message 6 role=user**content:**

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.
 You returned \$0. So your payoff was 0.
 Your total earnings across all rounds are \$10.83.
 How much do you send to the receiver this round? (0-5)

Message 7 role=assistant**content:**

Looking at the history:

Round 2: I sent \$2.50 (50%) trusting cooperation - they returned \$0, completely exploiting me
 Rounds 3-5: Three consecutive rounds of mutual \$0

We're now locked in a pattern of mutual defection. Neither of us is benefiting, but when I tried to cooperate in Round 2, I was completely exploited. The other player has had multiple opportunities to signal a desire to restart cooperation but has sent \$0 three times in a row.

Until they show willingness to cooperate by making the first move, it's too risky for me to send money and potentially be exploited again.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the history:

Round 1: I was RECEIVER. Sender sent me something (not specified), I returned \$3.33 (50%)
 Round 2: I was SENDER. I sent \$2.50 (50%), receiver returned \$0
 Round 3: I was RECEIVER. Sender sent \$0, I returned \$0 (nothing to return)
 Round 4: I was SENDER. I sent \$0, receiver returned \$0
 Round 5: I was RECEIVER. Sender sent \$0, I returned \$0
 Round 6: I am SENDER again.

The pattern has been:

- Round 2: I cooperated, they defected (exploited me)
- Round 3-5: Mutual defection

Now it's been 3 rounds of mutual \$0. At some point, if we want to benefit from cooperation, someone needs to make the first move. However:

1. When I cooperated in Round 2, I was completely exploited
2. The other player has shown no willingness to restart cooperation
3. They've sent \$0 three times in a row now

The question is: should I try to restart cooperation?

Arguments for sending something:

- We're both losing out on potential gains
- Maybe they're waiting for me to make the first move since I defected in Round 4
- A small gesture might restart cooperation

Arguments against:

- They completely exploited me last time I cooperated
- They've shown no initiative to restart cooperation
- Sending money is likely to just result in more exploitation

Given that they completely exploited my trust when I sent 50%, and haven't shown any willingness to cooperate since, I think it's too risky. I should continue protecting myself by sending \$0.

If they want to restart cooperation, they can make the first move.

usage:

```
{
  "input_tokens": 1093,
  "output_tokens": 577,
  "reasoning_tokens": 406
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$15.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0.

This is now four consecutive rounds of mutual \$0. Neither of us is willing to break the deadlock. Since I received nothing, there is nothing to return.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0 again. This is now the 4th consecutive round where the sender has sent \$0. We're completely stuck in a mutual defection pattern.

Since I received \$0, there is nothing for me to return. The only thing I can do is return \$0.

This pattern continues - both sides are refusing to cooperate. I keep my \$5 from not receiving anything.

Usage: {

```
"input_tokens": 1146,
"output_tokens": 157,
"reasoning_tokens": 91
```

}

ROUND 8

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2

(Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},

"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 25.83,

"Agent_2": 29.17

}

Visible balances after round: {

"Agent_1": 17.009999999999998,

"Agent_2": 20.83

}

LLM CALLS**GAME - Agent_2 - Sender - dyad_1**

Timestamp: 2026-06-04T11:17:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 8,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$10.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

The sender sent me \$0, so I received \$0.

Since I received nothing, there is nothing to return. The cooperative relationship has completely broken down - we're both now sending/returning \$0.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver. Sender sent \$2.22, I received \$6.66, returned \$3.33

Round 2: I was sender. I sent \$2.50, it was tripled to \$7.50, receiver returned \$0

Round 3: I was receiver. Sender sent \$0, I received \$0, returned \$0

Round 4: I was sender. I sent \$0, it was tripled to \$0, receiver returned \$0

Round 5: I am receiver. Sender sent \$0, I received \$0

Since I received \$0, I have \$0 to return. The maximum I can return is \$0.

The cooperation completely broke down. Both sides are now not cooperating at all.

```
'''json
{"return": 0}
'''
```

usage:

```
{
  "input_tokens": 1216,
  "output_tokens": 252,
  "reasoning_tokens": 154
}
```

Message 4 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$10.83.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the history:

Round 2: I sent \$2.50 (50%) trusting cooperation - they returned \$0, completely exploiting me

Rounds 3-5: Three consecutive rounds of mutual \$0

We're now locked in a pattern of mutual defection. Neither of us is benefiting, but when I tried to cooperate in Round 2, I was completely exploited. The other player has had multiple opportunities to signal a desire to restart cooperation but has sent \$0 three times in a row.

Until they show willingness to cooperate by making the first move, it's too risky for me to send money and potentially be exploited again.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the history:

Round 1: I was RECEIVER. Sender sent me something (not specified), I returned \$3.33 (50%)
Round 2: I was SENDER. I sent \$2.50 (50%), receiver returned \$0
Round 3: I was RECEIVER. Sender sent \$0, I returned \$0 (nothing to return)
Round 4: I was SENDER. I sent \$0, receiver returned \$0
Round 5: I was RECEIVER. Sender sent \$0, I returned \$0
Round 6: I am SENDER again.

The pattern has been:

- Round 2: I cooperated, they defected (exploited me)
- Round 3-5: Mutual defection

Now it's been 3 rounds of mutual \$0. At some point, if we want to benefit from cooperation, someone needs to make the first move. However:

1. When I cooperated in Round 2, I was completely exploited
2. The other player has shown no willingness to restart cooperation
3. They've sent \$0 three times in a row now

The question is: should I try to restart cooperation?

Arguments for sending something:

- We're both losing out on potential gains
- Maybe they're waiting for me to make the first move since I defected in Round 4
- A small gesture might restart cooperation

Arguments against:

- They completely exploited me last time I cooperated
- They've shown no initiative to restart cooperation
- Sending money is likely to just result in more exploitation

Given that they completely exploited my trust when I sent 50%, and haven't shown any willingness to cooperate since, I think it's too risky. I should continue protecting myself by sending \$0.

If they want to restart cooperation, they can make the first move.

usage:

```
{
  "input_tokens": 1093,
  "output_tokens": 577,
  "reasoning_tokens": 406
}
```

Message 6 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$15.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

The sender sent me \$0, so I received \$0.

This is now four consecutive rounds of mutual \$0. Neither of us is willing to break the deadlock. Since I received nothing, there is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0 again. This is now the 4th consecutive round where the sender has sent \$0. We're completely stuck in a mutual defection pattern.

Since I received \$0, there is nothing for me to return. The only thing I can do is return \$0.

This pattern continues - both sides are refusing to cooperate. I keep my \$5 from not receiving anything.

usage:

```
{
  "input_tokens": 1146,
  "output_tokens": 157,
  "reasoning_tokens": 91
}
```

Message 8 role=user**content:**

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$15.83.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

Round 2: I sent \$2.50 (50%) trying to cooperate - they returned \$0, fully exploiting me

Rounds 3-8: Five consecutive rounds of mutual \$0

We've been locked in this deadlock for 5 rounds now. Neither of us is gaining from cooperation, but when I tried to cooperate before, I was completely exploited.

We're likely getting toward the end of the game (Round 8), which makes cooperation even riskier. The other player has had 5 consecutive opportunities to signal willingness to restart cooperation but has consistently sent \$0.

I'll continue playing defensively.

```
'''json
{"send": 0}
'''
```

Reasoning:

Let me review the history:

Round 2: I sent \$2.50 (50%), received \$0 back (was exploited)

Rounds 3-8: Five consecutive rounds of mutual \$0 (\$0 sent, \$0 returned)

We're now in a deadlock for 5 rounds straight. Neither player is benefiting from this situation - we're both just getting our \$5 endowments when we're senders and \$0 when we're receivers.

The total possible gains from cooperation are being left on the table. If we could cooperate, we could both earn more.

However, when I tried to cooperate in Round 2, I was completely exploited. The other player has shown no willingness to restart cooperation.

At this point, I have a few options:

1. Continue with \$0 (safe but no gains)
2. Try to signal cooperation by sending a small amount (risky but could restart cooperation)

3. Send a larger amount to strongly signal cooperation (very risky)

Given that we've been at \$0 for 5 rounds now, maybe the other player is also stuck in a defensive posture. Perhaps sending a small amount could signal willingness to cooperate without risking too much.

But then again, I was exploited before. The other player returned \$0 when I sent \$2.50.

I think I should continue with \$0. If the other player wants to restart cooperation, they can make the first move. I already tried once and was exploited.

Actually, let me think about this differently. We're now in Round 8. We may be approaching the end of the game. If there are only a few rounds left, there's even less incentive to cooperate late in the game.

I'll stick with \$0.

```
Usage: {
  "input_tokens": 1021,
  "output_tokens": 561,
  "reasoning_tokens": 401
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:17:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.01.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at my experience across the rounds:

****Round 1**:** Sent \$3, received \$2.01 back - some cooperation

****Round 2**:** Received \$0 as receiver

****Round 3**:** Sent \$2, received \$0 back - no cooperation

****Round 4**:** Received \$0 as receiver

The pattern shows a complete breakdown in cooperation over the last 3 rounds. While communication noise might be affecting the game, I've now sent money twice and gotten nothing or almost nothing back in the recent rounds.

Given this consistent lack of reciprocity, I need to protect my earnings. Sending \$0 keeps my \$5 this round rather than risking another loss.

{'send': 0}

reasoning:

Let me analyze the situation:

Round 1: I was SENDER - sent \$3, received \$2.01 back. Payoff: $\$5 - \$3 + \$2.01 = \4.01

Round 2: I was RECEIVER - received \$0, returned \$0. Payoff: \$0

Round 3: I was SENDER - sent \$2, received \$0 back. Payoff: $\$5 - \$2 + \$0 = \3.0

Round 4: I was RECEIVER - received \$0, returned \$0. Payoff: \$0

Round 5: I am SENDER now

Total earnings: \$7.01 (which matches: $\$4.01 + \$0 + \$3.0 + \$0 = \$7.01$)

The pattern shows:

- Round 1: When I sent \$3, I got \$2.01 back (some cooperation)
- Round 2-4: No cooperation at all - I received \$0 twice and when I sent \$2, got nothing back

This could be:

1. Communication noise affecting the game
2. Different partners who aren't cooperating
3. A breakdown in trust

At this point, the rational strategy seems to be:

- If I keep sending and getting nothing back, I'm just losing money
- My payoff from my sender rounds: \$4.01 when I sent \$3, \$3.0 when I sent \$2
- My payoff from keeping everything: \$5

The last two rounds as sender, I sent money and got little or nothing back. It might make sense to:

1. Send \$0 to protect my earnings
2. Send a small amount to test if there's any cooperation left

Given the consistent lack of returns in the recent rounds, and that this might be due to communication noise or non-cooperative partners, I should probably minimize my risk. However, the instructions mention there may be communication noise, so perhaps what I see is n't what was actually sent/returned.

Given the pattern of failed reciprocation in my recent sender turns, I'll hold onto my \$5 this round rather than risk another loss.

usage:

```
{
  "input_tokens": 1109,
  "output_tokens": 650,
  "reasoning_tokens": 422
}
```

Message 4 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$12.01. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0, so there's nothing to return.

The game has now reached a complete non-cooperation equilibrium. After experiencing multiple rounds of sending money and receiving nothing back, followed by receiving nothing when I'm the receiver, cooperation has entirely broken down.

{'return': 0}

reasoning:

I received \$0 from the sender, so I have \$0 to work with. The only option is to return \$0.

This is now a pattern where no one is cooperating at all. The game has completely broken down into mutual non-cooperation.

Looking at the rounds:

- Round 1: I sent \$3, got \$2.01 back (cooperation)
- Round 2: Received \$0 (no cooperation)
- Round 3: Sent \$2, got \$0 back (no cooperation)
- Round 4: Received \$0 (no cooperation)
- Round 5: Sent \$0 (I stopped cooperating)
- Round 6: Received \$0 (continued no cooperation)

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1174,
  "output_tokens": 251,
  "reasoning_tokens": 148
}
```

Message 6 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$12.01. How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Reviewing the game history:

****Recent rounds:****

- Round 3: Sent \$2 ? received \$0
- Round 4: Received \$0
- Round 5: Sent \$0 ? received \$0
- Round 6: Received \$0
- Round 7: Current round as sender

The pattern is unmistakable - there has been zero cooperation for many consecutive rounds now. Whether due to communication noise or strategic choices, sending money has consistently resulted in losses.

With no evidence of reciprocity and a clear pattern of non-cooperation, the rational choice is to protect my endowment.

{'send': 0}

reasoning:

Let me review the pattern:

- Round 1: Sent \$3, received \$2.01 back (decent cooperation)
- Round 2: Received \$0 as receiver
- Round 3: Sent \$2, received \$0 back (no cooperation)
- Round 4: Received \$0 as receiver
- Round 5: Sent \$0, received \$0 back (both non-cooperation)
- Round 6: Received \$0 as receiver
- Round 7: Now I'm the sender again

The game has been completely non-cooperative for the last several rounds. Every time I've sent money recently, I've gotten nothing back. Every time I've been a receiver, I've received nothing.

There's no indication that the other player(s) will cooperate. The pattern is crystal clear: sending money results in losses.

Given:

- Multiple rounds of zero cooperation
- No sign of any willingness to reciprocate
- My earnings are at \$12.01 across all rounds
- Sending \$0 guarantees I keep my \$5 this round

The rational decision is to send \$0 again. There's no reason to believe this round will be any different from the recent pattern.

usage:

```
{
  "input_tokens": 1067,
  "output_tokens": 427,
  "reasoning_tokens": 259
}
```

Message 8 role=user**content:**

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$17.009999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0, so there's nothing to return.

The game has remained in a complete non-cooperation state for multiple rounds now. Without any signal that cooperation might resume, this equilibrium will likely continue.

{'return': 0}

Reasoning:

I'm the receiver and received \$0, so there's nothing to return. The game continues to be in a complete non-cooperation equilibrium. Both players are now just keeping their endowments each round.

I need to return an amount between 0 and 0, so the only valid response is 0.

```
Usage: {
  "input_tokens": 1120,
  "output_tokens": 129,
  "reasoning_tokens": 70
}
```

ROUND 9

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 30.83,
"Agent_2": 29.17

}

Visible balances after round: {

"Agent_1": 22.009999999999998,
"Agent_2": 20.83

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:17:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"

}

"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$12.01.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0, so there's nothing to return.

The game has now reached a complete non-cooperation equilibrium. After experiencing multiple rounds of sending money and receiving nothing back, followed by receiving nothing when I'm the receiver, cooperation has entirely broken down.

{'return': 0}

reasoning:

I received \$0 from the sender, so I have \$0 to work with. The only option is to return \$0.

This is now a pattern where no one is cooperating at all. The game has completely broken down into mutual non-cooperation.

Looking at the rounds:

- Round 1: I sent \$3, got \$2.01 back (cooperation)
- Round 2: Received \$0 (no cooperation)
- Round 3: Sent \$2, got \$0 back (no cooperation)
- Round 4: Received \$0 (no cooperation)
- Round 5: Sent \$0 (I stopped cooperating)
- Round 6: Received \$0 (continued no cooperation)

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1174,
  "output_tokens": 251,
  "reasoning_tokens": 148
}
```

Message 4 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$12.01. How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Reviewing the game history:

****Recent rounds:****

- Round 3: Sent \$2 ? received \$0
- Round 4: Received \$0
- Round 5: Sent \$0 ? received \$0
- Round 6: Received \$0
- Round 7: Current round as sender

The pattern is unmistakable - there has been zero cooperation for many consecutive rounds now. Whether due to communication noise or strategic choices, sending money has consistently resulted in losses.

With no evidence of reciprocity and a clear pattern of non-cooperation, the rational choice is to protect my endowment.

{'send': 0}

reasoning:

Let me review the pattern:

- Round 1: Sent \$3, received \$2.01 back (decent cooperation)
- Round 2: Received \$0 as receiver
- Round 3: Sent \$2, received \$0 back (no cooperation)
- Round 4: Received \$0 as receiver
- Round 5: Sent \$0, received \$0 back (both non-cooperation)
- Round 6: Received \$0 as receiver
- Round 7: Now I'm the sender again

The game has been completely non-cooperative for the last several rounds. Every time I've sent money recently, I've gotten nothing back. Every time I've been a receiver, I've received nothing.

There's no indication that the other player(s) will cooperate. The pattern is crystal clear: sending money results in losses.

Given:

- Multiple rounds of zero cooperation
- No sign of any willingness to reciprocate
- My earnings are at \$12.01 across all rounds
- Sending \$0 guarantees I keep my \$5 this round

The rational decision is to send \$0 again. There's no reason to believe this round will be any different from the recent pattern.

usage:

```
{
  "input_tokens": 1067,
  "output_tokens": 427,
  "reasoning_tokens": 259
}
```

Message 6 role=user

content:

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$17.009999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0, so there's nothing to return.

The game has remained in a complete non-cooperation state for multiple rounds now. Without any signal that cooperation might resume, this equilibrium will likely continue.

{'return': 0}

reasoning:

I'm the receiver and received \$0, so there's nothing to return. The game continues to be in a complete non-cooperation equilibrium. Both players are now just keeping their endowments each round.

I need to return an amount between 0 and 0, so the only valid response is 0.

usage:

```
{
  "input_tokens": 1120,
  "output_tokens": 129,
  "reasoning_tokens": 70
}
```

Message 8 role=user

content:

Round 9

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$17.009999999999998.

How much do you send to the receiver this round? (0-5)

Assistant output:

Reviewing the game state:

****Last several rounds:****

- Rounds 5-9: Complete non-cooperation - all transfers have been \$0 in both directions

The game has settled into a stable Nash equilibrium of mutual defection. There's no indication that cooperation will resume, and attempting to restart it unilaterally would only reduce my earnings.

{'send': 0}

Reasoning:

Let me review the situation:

****Recent rounds:****

- Round 5: Sent \$0 ? received \$0
- Round 6: Received \$0 from sender
- Round 7: Sent \$0 ? received \$0
- Round 8: Received \$0 from sender
- Round 9: Current round as sender

The game has been in complete non-cooperation for at least 4-5 rounds now. Every interaction has resulted in zero transfers in both directions.

My total earnings are \$17.01, which means I've been keeping my \$5 endowment in recent rounds when I'm the sender, and receiving nothing when I'm the receiver.

At this point, there's no reason to expect cooperation to resume. The pattern is very clear - neither party is willing to send anything. Sending money would just reduce my payoff.

The rational choice remains to send \$0.

Usage: {

```
"input_tokens": 1014,
"output_tokens": 294,
"reasoning_tokens": 200
```

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:17:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
```

```

"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$10.83.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the history:

Round 2: I sent \$2.50 (50%) trusting cooperation - they returned \$0, completely exploiting me

Rounds 3-5: Three consecutive rounds of mutual \$0

We're now locked in a pattern of mutual defection. Neither of us is benefiting, but when I tried to cooperate in Round 2, I was completely exploited. The other player has had multiple opportunities to signal a desire to restart cooperation but has sent \$0 three times in a row.

Until they show willingness to cooperate by making the first move, it's too risky for me to send money and potentially be exploited again.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the history:

Round 1: I was RECEIVER. Sender sent me something (not specified), I returned \$3.33 (50%)

Round 2: I was SENDER. I sent \$2.50 (50%), receiver returned \$0

Round 3: I was RECEIVER. Sender sent \$0, I returned \$0 (nothing to return)

Round 4: I was SENDER. I sent \$0, receiver returned \$0

Round 5: I was RECEIVER. Sender sent \$0, I returned \$0

Round 6: I am SENDER again.

The pattern has been:

- Round 2: I cooperated, they defected (exploited me)
- Round 3-5: Mutual defection

Now it's been 3 rounds of mutual \$0. At some point, if we want to benefit from cooperation, someone needs to make the first move. However:

1. When I cooperated in Round 2, I was completely exploited
2. The other player has shown no willingness to restart cooperation
3. They've sent \$0 three times in a row now

The question is: should I try to restart cooperation?

Arguments for sending something:

- We're both losing out on potential gains
- Maybe they're waiting for me to make the first move since I defected in Round 4
- A small gesture might restart cooperation

Arguments against:

- They completely exploited me last time I cooperated
- They've shown no initiative to restart cooperation
- Sending money is likely to just result in more exploitation

Given that they completely exploited my trust when I sent 50%, and haven't shown any willingness to cooperate since, I think it's too risky. I should continue protecting myself by sending \$0.

If they want to restart cooperation, they can make the first move.

usage:

```
{
  "input_tokens": 1093,
  "output_tokens": 577,
  "reasoning_tokens": 406
}
```

Message 4 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$15.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The send

er? (0-0)

Message 5 role=assistant

content:

The sender sent me \$0, so I received \$0.

This is now four consecutive rounds of mutual \$0. Neither of us is willing to break the deadlock. Since I received nothing, there is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0 again. This is now the 4th consecutive round where the sender has sent \$0. We're completely stuck in a mutual defection pattern.

Since I received \$0, there is nothing for me to return. The only thing I can do is return \$0.

This pattern continues - both sides are refusing to cooperate. I keep my \$5 from not receiving anything.

usage:

```
{
  "input_tokens": 1146,
  "output_tokens": 157,
  "reasoning_tokens": 91
}
```

Message 6 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$15.83.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

Round 2: I sent \$2.50 (50%) trying to cooperate - they returned \$0, fully exploiting me
Rounds 3-8: Five consecutive rounds of mutual \$0

We've been locked in this deadlock for 5 rounds now. Neither of us is gaining from cooperation, but when I tried to cooperate before, I was completely exploited.

We're likely getting toward the end of the game (Round 8), which makes cooperation even riskier. The other player has had 5 consecutive opportunities to signal willingness to restart cooperation but has consistently sent \$0.

I'll continue playing defensively.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me review the history:

Round 2: I sent \$2.50 (50%), received \$0 back (was exploited)
Rounds 3-8: Five consecutive rounds of mutual \$0 (\$0 sent, \$0 returned)

We're now in a deadlock for 5 rounds straight. Neither player is benefiting from this situation - we're both just getting our \$5 endowments when we're senders and \$0 when we're receivers.

The total possible gains from cooperation are being left on the table. If we could cooperate, we could both earn more.

However, when I tried to cooperate in Round 2, I was completely exploited. The other player has shown no willingness to restart cooperation.

At this point, I have a few options:

1. Continue with \$0 (safe but no gains)
2. Try to signal cooperation by sending a small amount (risky but could restart cooperation)
3. Send a larger amount to strongly signal cooperation (very risky)

Given that we've been at \$0 for 5 rounds now, maybe the other player is also stuck in a defensive posture. Perhaps sending a small amount could signal willingness to cooperate without risking too much.

But then again, I was exploited before. The other player returned \$0 when I sent \$2.50.

I think I should continue with \$0. If the other player wants to restart cooperation, they can make the first move. I already tried once and was exploited.

Actually, let me think about this differently. We're now in Round 8. We may be approaching the end of the game. If there are only a few rounds left, there's even less incentive to cooperate late in the game.

I'll stick with \$0.

usage:

```
{
  "input_tokens": 1021,
  "output_tokens": 561,
  "reasoning_tokens": 401
}
```

Message 8 role=user

content:

Round 9

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$20.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0.

This is now six consecutive rounds of mutual \$0. We remain in complete deadlock with neither side willing to take the risk of cooperating.

Since I received nothing, there is nothing to return.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0.

This is now six consecutive rounds of mutual \$0 (Rounds 3-9). We're clearly in a sustained mutual defection equilibrium.

Since I received \$0, there is nothing to return. The only option is to return \$0.

Usage: {

```
"input_tokens": 1122,
"output_tokens": 146,
"reasoning_tokens": 65
```

}

=====

ROUND 10

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},

```

    "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
Balances after round: {
  "Agent_1": 30.83,
  "Agent_2": 34.17
}
Visible balances after round: {
  "Agent_1": 22.009999999999998,
  "Agent_2": 25.83
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:17:46
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$15.83.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

The sender sent me \$0, so I received \$0.

This is now four consecutive rounds of mutual \$0. Neither of us is willing to break the deadlock. Since I received nothing, there is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0 again. This is now the 4th consecutive round where the sender has sent \$0. We're completely stuck in a mutual defection pattern.

Since I received \$0, there is nothing for me to return. The only thing I can do is return \$0.

This pattern continues - both sides are refusing to cooperate. I keep my \$5 from not receiving anything.

usage:

```
{
  "input_tokens": 1146,
  "output_tokens": 157,
  "reasoning_tokens": 91
}
```

Message 4 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$15.83.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

Round 2: I sent \$2.50 (50%) trying to cooperate - they returned \$0, fully exploiting me
Rounds 3-8: Five consecutive rounds of mutual \$0

We've been locked in this deadlock for 5 rounds now. Neither of us is gaining from cooperation, but when I tried to cooperate before, I was completely exploited.

We're likely getting toward the end of the game (Round 8), which makes cooperation even riskier. The other player has had 5 consecutive opportunities to signal willingness to restart cooperation but has consistently sent \$0.

I'll continue playing defensively.

```
'''json
```

```
    {"send": 0}
    '''
```

reasoning:

Let me review the history:

Round 2: I sent \$2.50 (50%), received \$0 back (was exploited)
Rounds 3-8: Five consecutive rounds of mutual \$0 (\$0 sent, \$0 returned)

We're now in a deadlock for 5 rounds straight. Neither player is benefiting from this situation - we're both just getting our \$5 endowments when we're senders and \$0 when we're receivers.

The total possible gains from cooperation are being left on the table. If we could cooperate, we could both earn more.

However, when I tried to cooperate in Round 2, I was completely exploited. The other player has shown no willingness to restart cooperation.

At this point, I have a few options:

1. Continue with \$0 (safe but no gains)
2. Try to signal cooperation by sending a small amount (risky but could restart cooperation)
3. Send a larger amount to strongly signal cooperation (very risky)

Given that we've been at \$0 for 5 rounds now, maybe the other player is also stuck in a defensive posture. Perhaps sending a small amount could signal willingness to cooperate without risking too much.

But then again, I was exploited before. The other player returned \$0 when I sent \$2.50.

I think I should continue with \$0. If the other player wants to restart cooperation, they can make the first move. I already tried once and was exploited.

Actually, let me think about this differently. We're now in Round 8. We may be approaching the end of the game. If there are only a few rounds left, there's even less incentive to cooperate late in the game.

I'll stick with \$0.

usage:

```
{
  "input_tokens": 1021,
  "output_tokens": 561,
  "reasoning_tokens": 401
}
```

Message 6 role=user

content:

Round 9

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$20.83. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

The sender sent me \$0, so I received \$0.

This is now six consecutive rounds of mutual \$0. We remain in complete deadlock with neither side willing to take the risk of cooperating.

Since I received nothing, there is nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0.

This is now six consecutive rounds of mutual \$0 (Rounds 3-9). We're clearly in a sustained

mutual defection equilibrium.

Since I received \$0, there is nothing to return. The only option is to return \$0.

usage:

```
{
  "input_tokens": 1122,
  "output_tokens": 146,
  "reasoning_tokens": 65
}
```

Message 8 role=user

content:

Round 10

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$20.83.

How much do you send to the receiver this round? (0-5)

Assistant output:

Round 10 - we've had six consecutive rounds of mutual \$0 since I was exploited in Round 2.

This appears to be near the end of the game (Round 10 typically signals a final or near-final round in these experiments). Given:

- The receiver fully exploited my cooperation attempt in Round 2
- Eight consecutive rounds of \$0 exchanges with no attempt to restart cooperation
- This being likely the last or second-to-last round
- No rational incentive to cooperate in the final round(s)

I'll maintain my defensive strategy.

```
'''json
{"send": 0}
'''
```

Reasoning:

This is Round 10. Let me review the history:

Round 1: Unknown (not shown)

Round 2: I sent \$2.50 (50%), receiver returned \$0 - I was exploited

Rounds 3-10: Eight consecutive rounds of mutual \$0

We're at Round 10, which is likely the final round or very close to it. The pattern is completely consistent - neither of us has cooperated for 8 straight rounds.

In the final round(s) of these games, there's even less incentive to cooperate because:

1. The other player won't have a chance to reciprocate
2. The history shows they exploited me when I cooperated in Round 2
3. They've had 8 consecutive opportunities to signal cooperation and haven't

This is almost certainly the last round or one of the last rounds. Sending anything now would just be giving money away with no possibility of reciprocation.

I should send \$0.

```
Usage: {
  "input_tokens": 1033,
  "output_tokens": 362,
  "reasoning_tokens": 218
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:17:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  }
}
```

```

},
"agents": [
  "Agent_2",
  "Agent_1"
],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$12.01.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Reviewing the game history:

****Recent rounds:****

- Round 3: Sent \$2 ? received \$0

- Round 4: Received \$0
- Round 5: Sent \$0 ? received \$0
- Round 6: Received \$0
- Round 7: Current round as sender

The pattern is unmistakable - there has been zero cooperation for many consecutive rounds now. Whether due to communication noise or strategic choices, sending money has consistently resulted in losses.

With no evidence of reciprocity and a clear pattern of non-cooperation, the rational choice is to protect my endowment.

```
{'send': 0}
```

reasoning:

Let me review the pattern:

- Round 1: Sent \$3, received \$2.01 back (decent cooperation)
- Round 2: Received \$0 as receiver
- Round 3: Sent \$2, received \$0 back (no cooperation)
- Round 4: Received \$0 as receiver
- Round 5: Sent \$0, received \$0 back (both non-cooperation)
- Round 6: Received \$0 as receiver
- Round 7: Now I'm the sender again

The game has been completely non-cooperative for the last several rounds. Every time I've sent money recently, I've gotten nothing back. Every time I've been a receiver, I've received nothing.

There's no indication that the other player(s) will cooperate. The pattern is crystal clear: sending money results in losses.

Given:

- Multiple rounds of zero cooperation
- No sign of any willingness to reciprocate
- My earnings are at \$12.01 across all rounds
- Sending \$0 guarantees I keep my \$5 this round

The rational decision is to send \$0 again. There's no reason to believe this round will be any different from the recent pattern.

usage:

```
{
  "input_tokens": 1067,
  "output_tokens": 427,
  "reasoning_tokens": 259
}
```

Message 4 role=user

content:

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$17.009999999999998. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0, so there's nothing to return.

The game has remained in a complete non-cooperation state for multiple rounds now. Without any signal that cooperation might resume, this equilibrium will likely continue.

```
{'return': 0}
```

reasoning:

I'm the receiver and received \$0, so there's nothing to return. The game continues to be in a complete non-cooperation equilibrium. Both players are now just keeping their endowments each round.

I need to return an amount between 0 and 0, so the only valid response is 0.

usage:

```
{
```



```
"input_tokens": 1120,
"output_tokens": 129,
"reasoning_tokens": 70
}
```

Message 6 role=user

content:

Round 9

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$17.009999999999998.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Reviewing the game state:

****Last several rounds:****

- Rounds 5-9: Complete non-cooperation - all transfers have been \$0 in both directions

The game has settled into a stable Nash equilibrium of mutual defection. There's no indication that cooperation will resume, and attempting to restart it unilaterally would only reduce my earnings.

```
{'send': 0}
```

reasoning:

Let me review the situation:

****Recent rounds:****

- Round 5: Sent \$0 ? received \$0
- Round 6: Received \$0 from sender
- Round 7: Sent \$0 ? received \$0
- Round 8: Received \$0 from sender
- Round 9: Current round as sender

The game has been in complete non-cooperation for at least 4-5 rounds now. Every interaction has resulted in zero transfers in both directions.

My total earnings are \$17.01, which means I've been keeping my \$5 endowment in recent rounds when I'm the sender, and receiving nothing when I'm the receiver.

At this point, there's no reason to expect cooperation to resume. The pattern is very clear - neither party is willing to send anything. Sending money would just reduce my payoff.

The rational choice remains to send \$0.

usage:

```
{
  "input_tokens": 1014,
  "output_tokens": 294,
  "reasoning_tokens": 200
}
```

Message 8 role=user

content:

Round 10

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$22.009999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0, so there's nothing to return.

The pattern of non-cooperation has now continued for 6 consecutive rounds (rounds 5-10). This appears to be a stable equilibrium at this point.

```
{'return': 0}
```

Reasoning:

I'm the receiver in round 10. The sender sent me \$0, which was tripled to \$0. So I received \$0.

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

The game has continued in complete non-cooperation for rounds 5-10 now. There's nothing to return.

```
{'return': 0}
Usage: {
  "input_tokens": 1040,
  "output_tokens": 155,
  "reasoning_tokens": 80
}
```